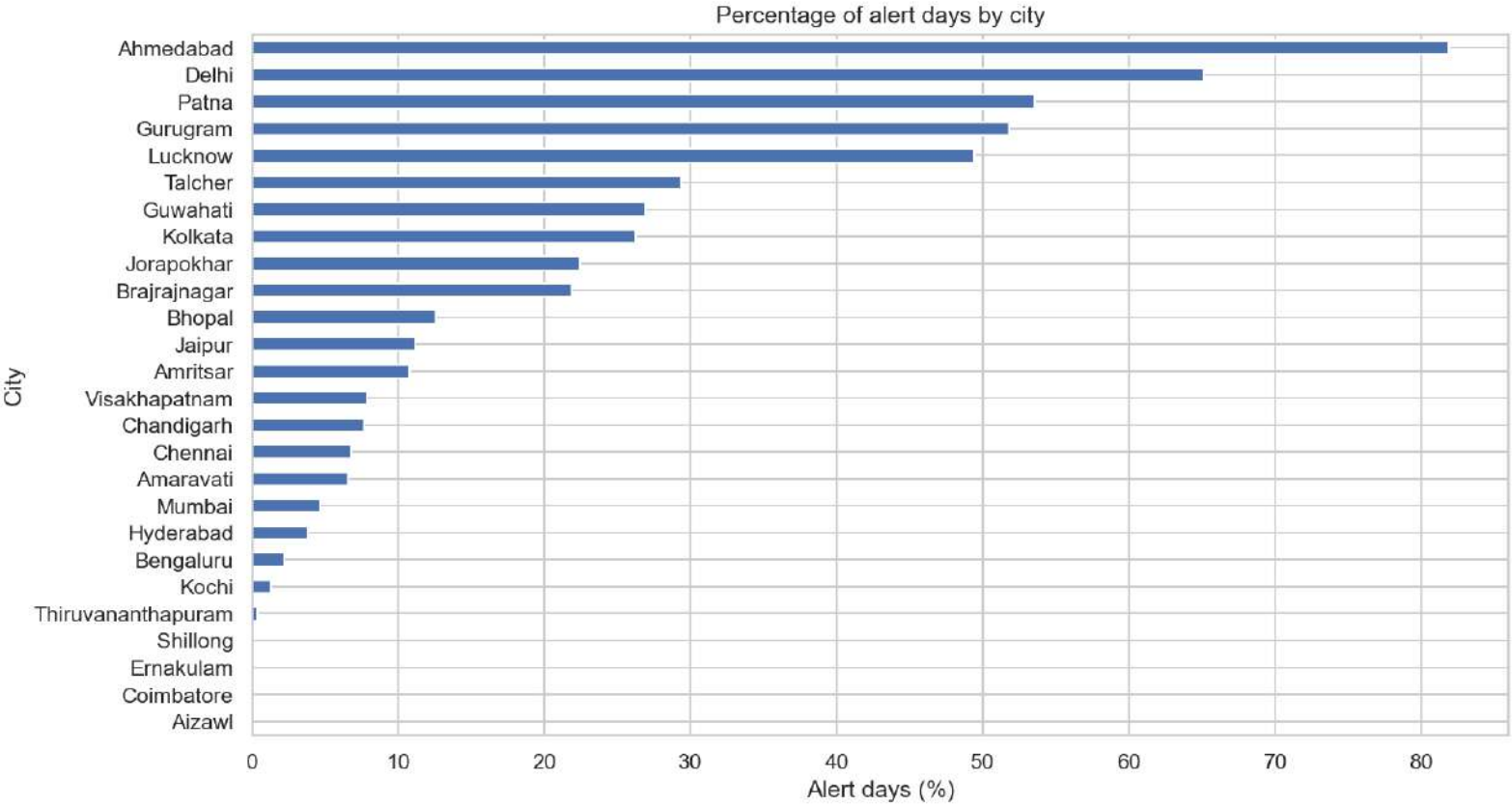
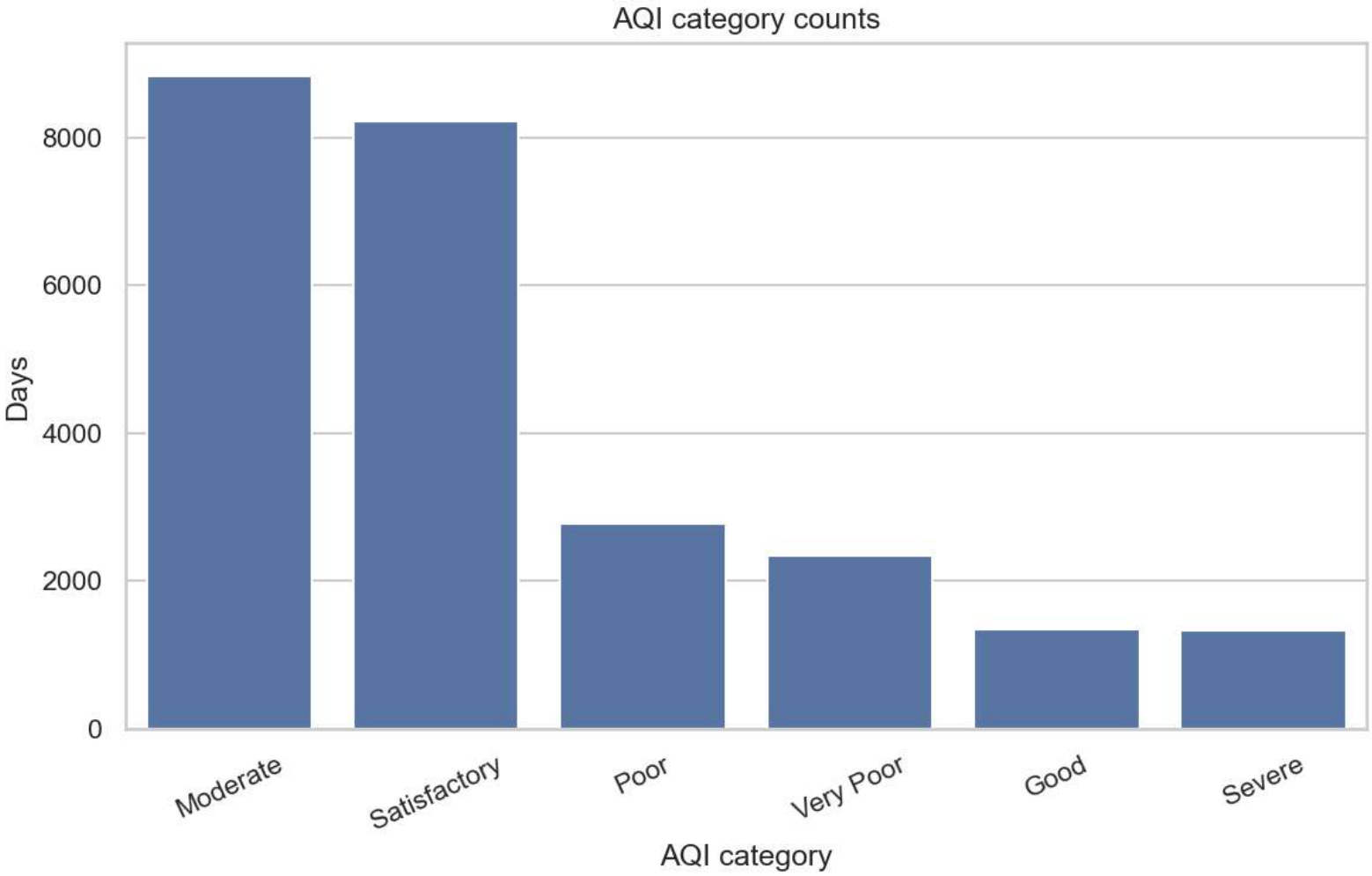
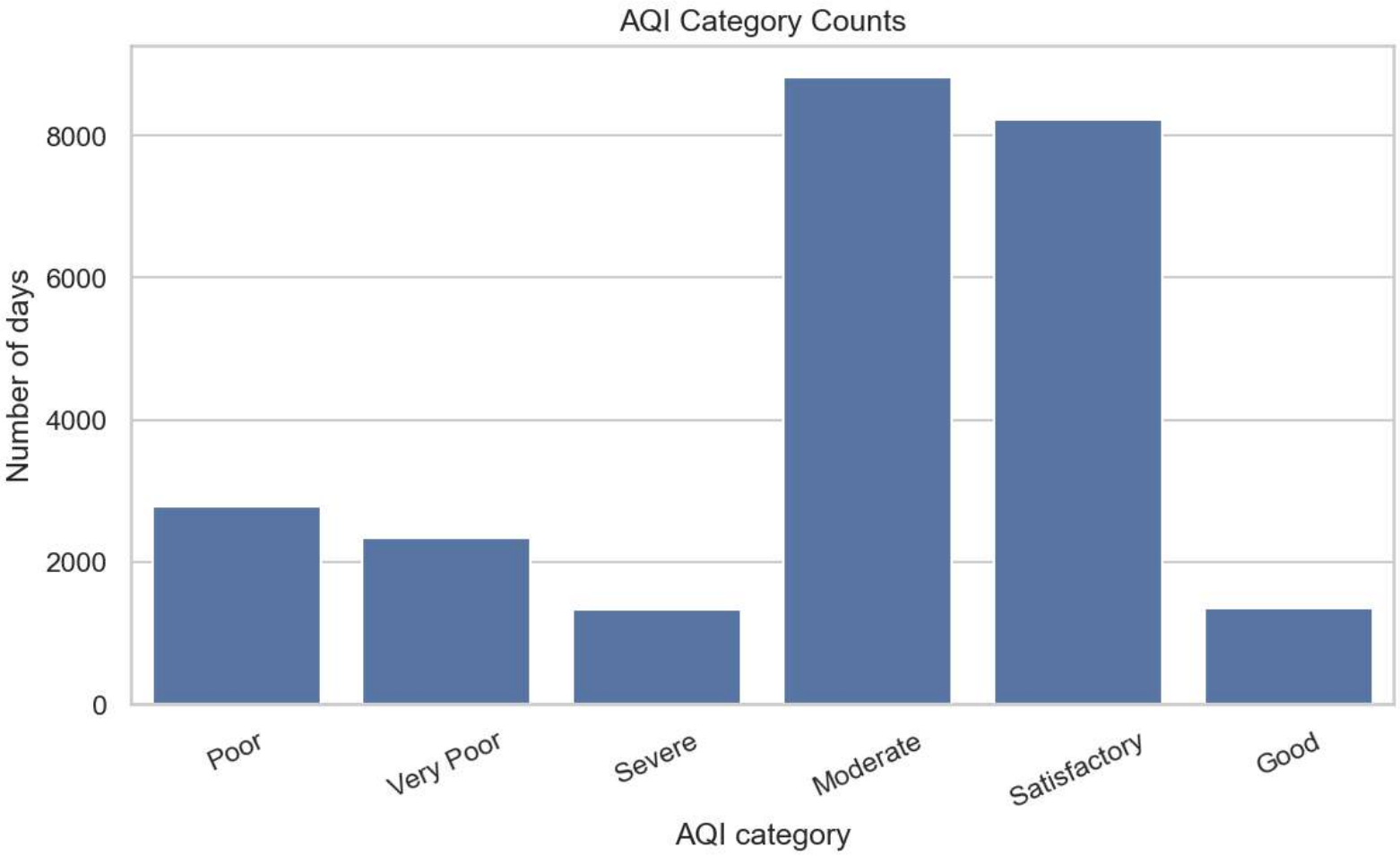
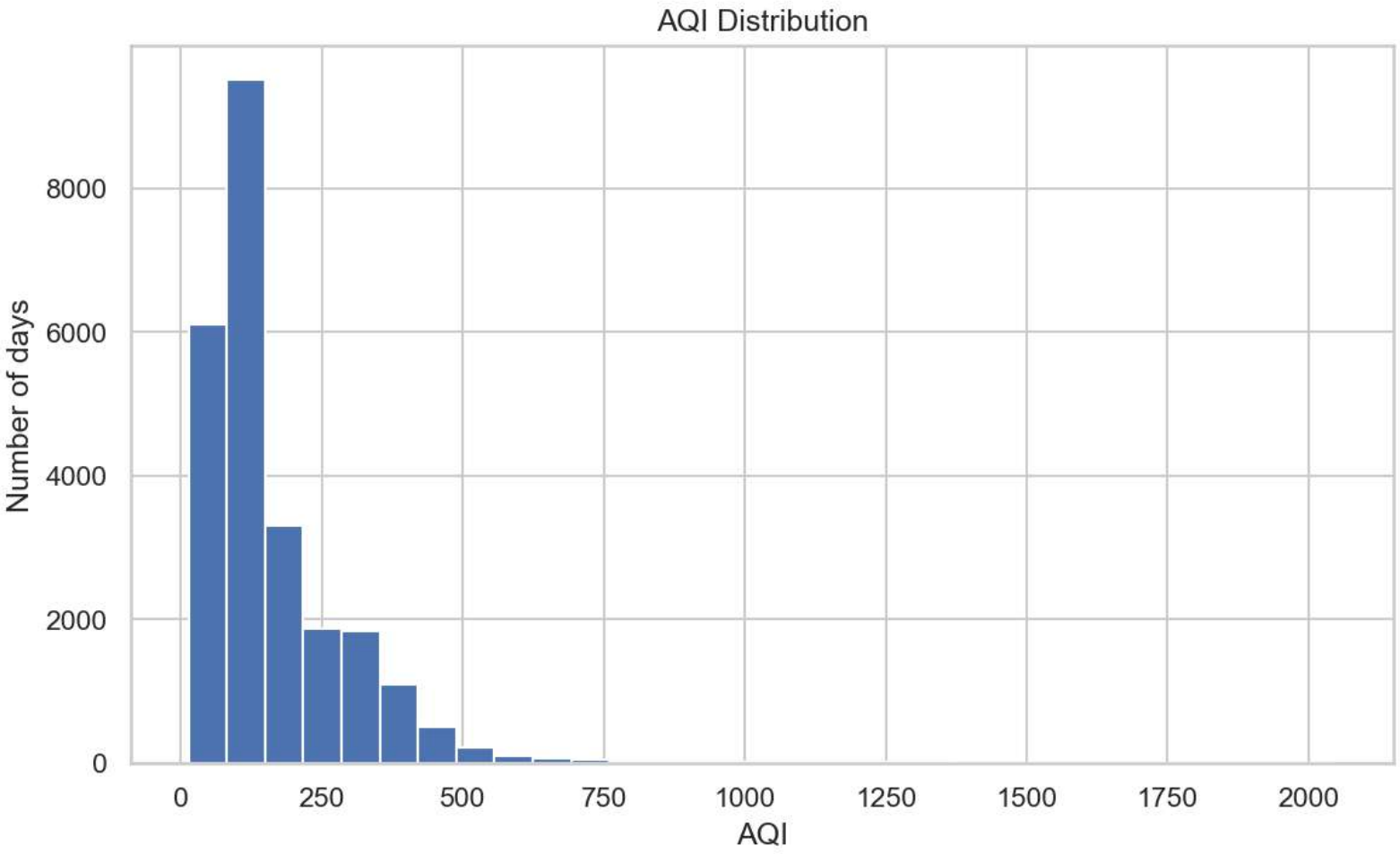
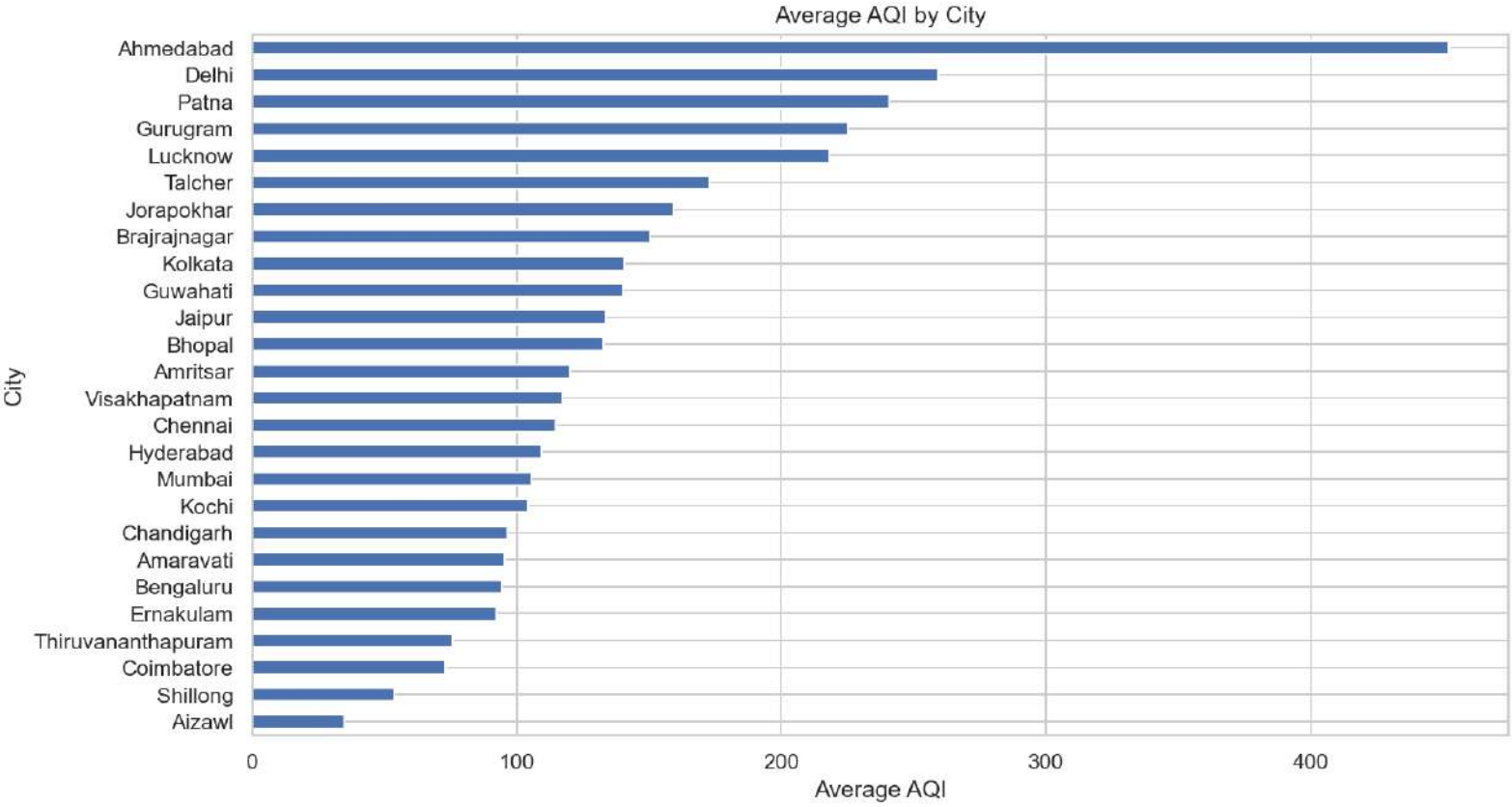
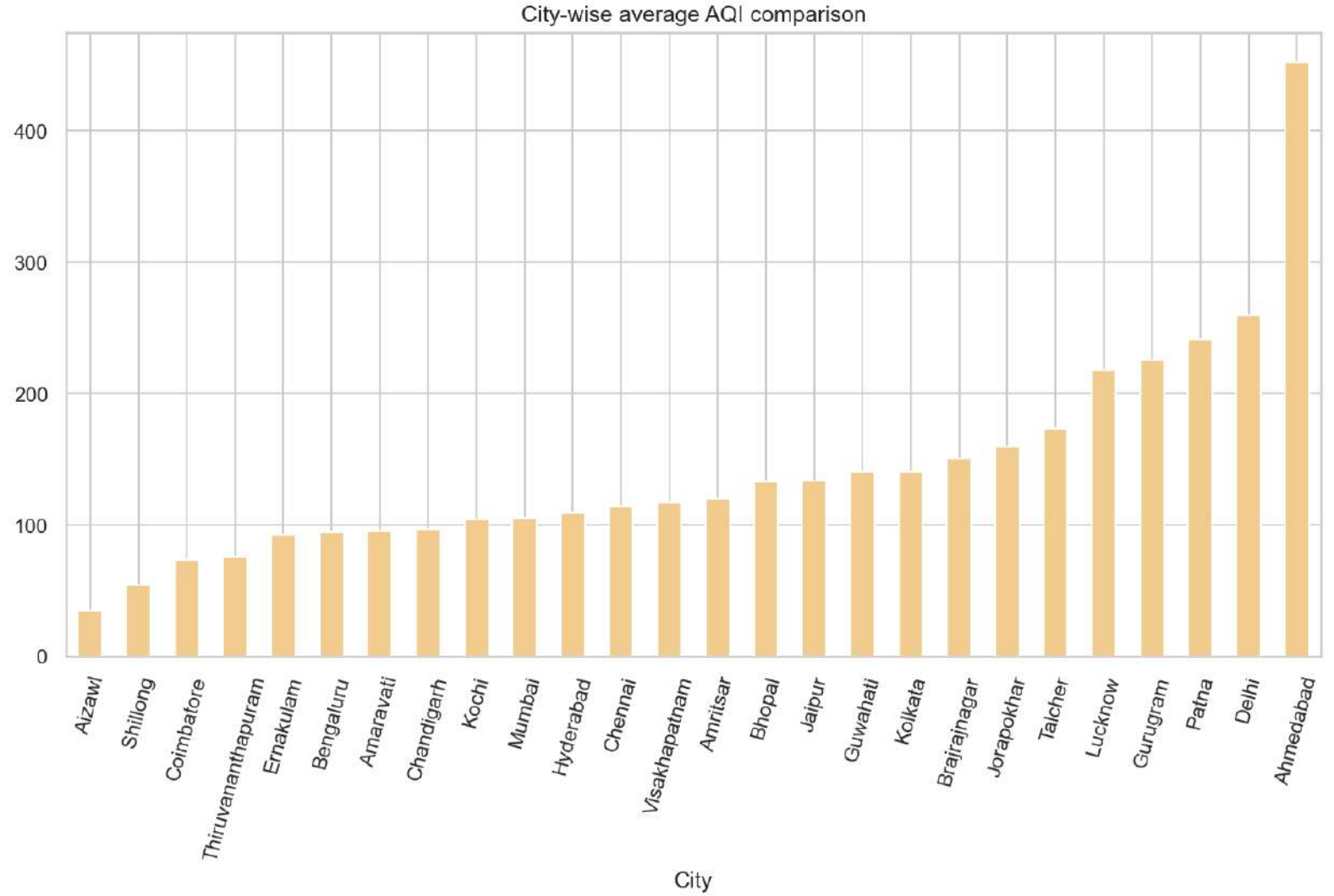

Percentage of alert days by city
Ahmedabad
Delhi
Patna
Gurugram
Lucknow
Talcher
Guwahati
Kolkata
Jorapokhar
Brajrajnagar
Bhopal
Jaipur
Amritsar
Visakhapatnam
Chandigarh
Chennai
Amaravati
Mumbai
Hyderabad
Bengaluru
Kochi
Thiruvananthapuram
Shillong
Ernakulam
Coimbatore
Aizawl
City
0
10
20
30
40
50
60
70
80
Alert days (%)

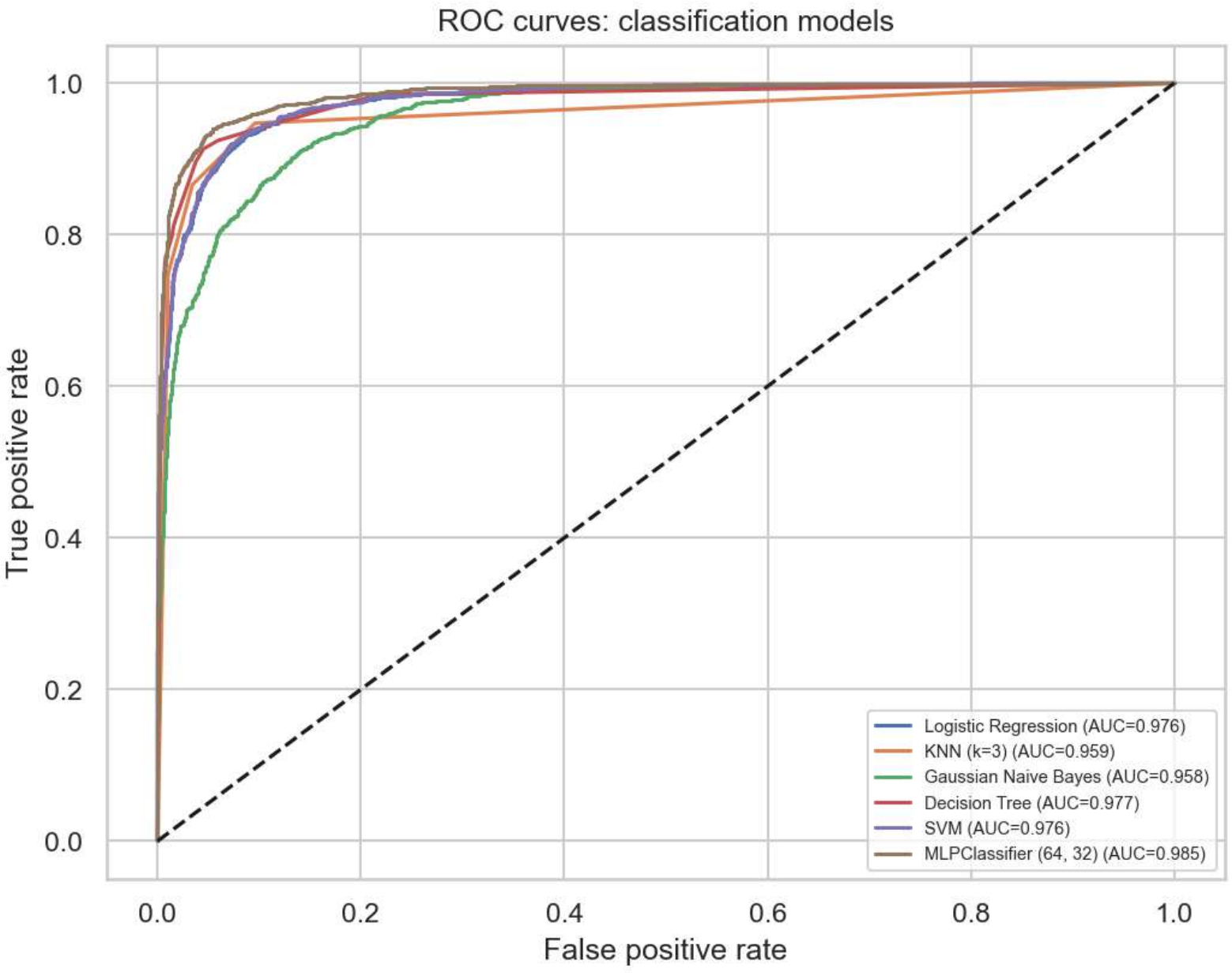

AQI category counts
Days
8000
6000
4000
2000
0
Moderate
Satisfactory
Poor
Very Poor
Good
Severe
AQI category

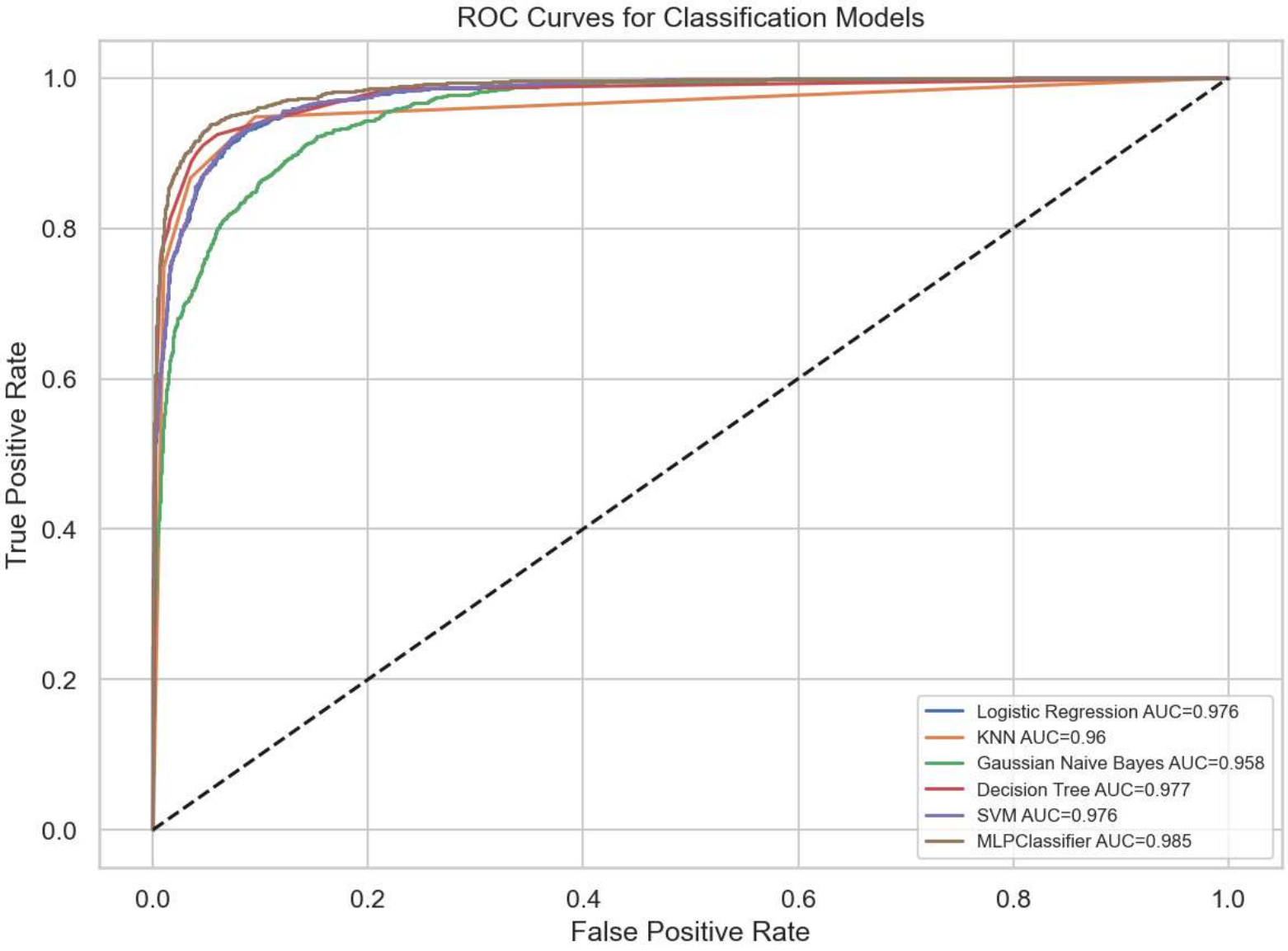

AQI Category Counts
Number of days
0
2000
4000
6000
8000
Poor
Very Poor
Severe
Moderate
Satisfactory
Good
AQI category

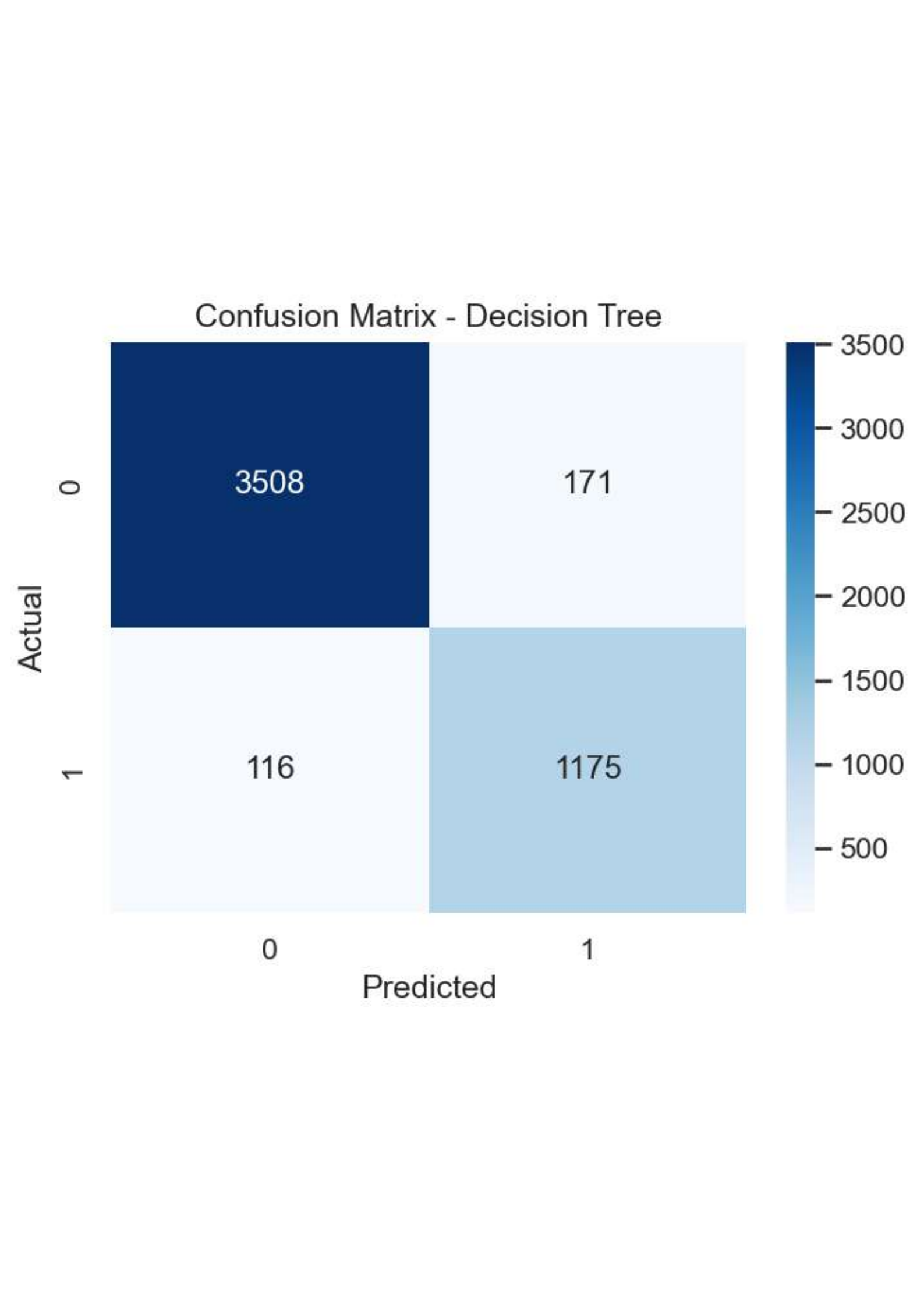

AQI Distribution
Number of days
8000
6000
4000
2000
0
0
250
500
750
1000
1250
1500
1750
2000
AQI

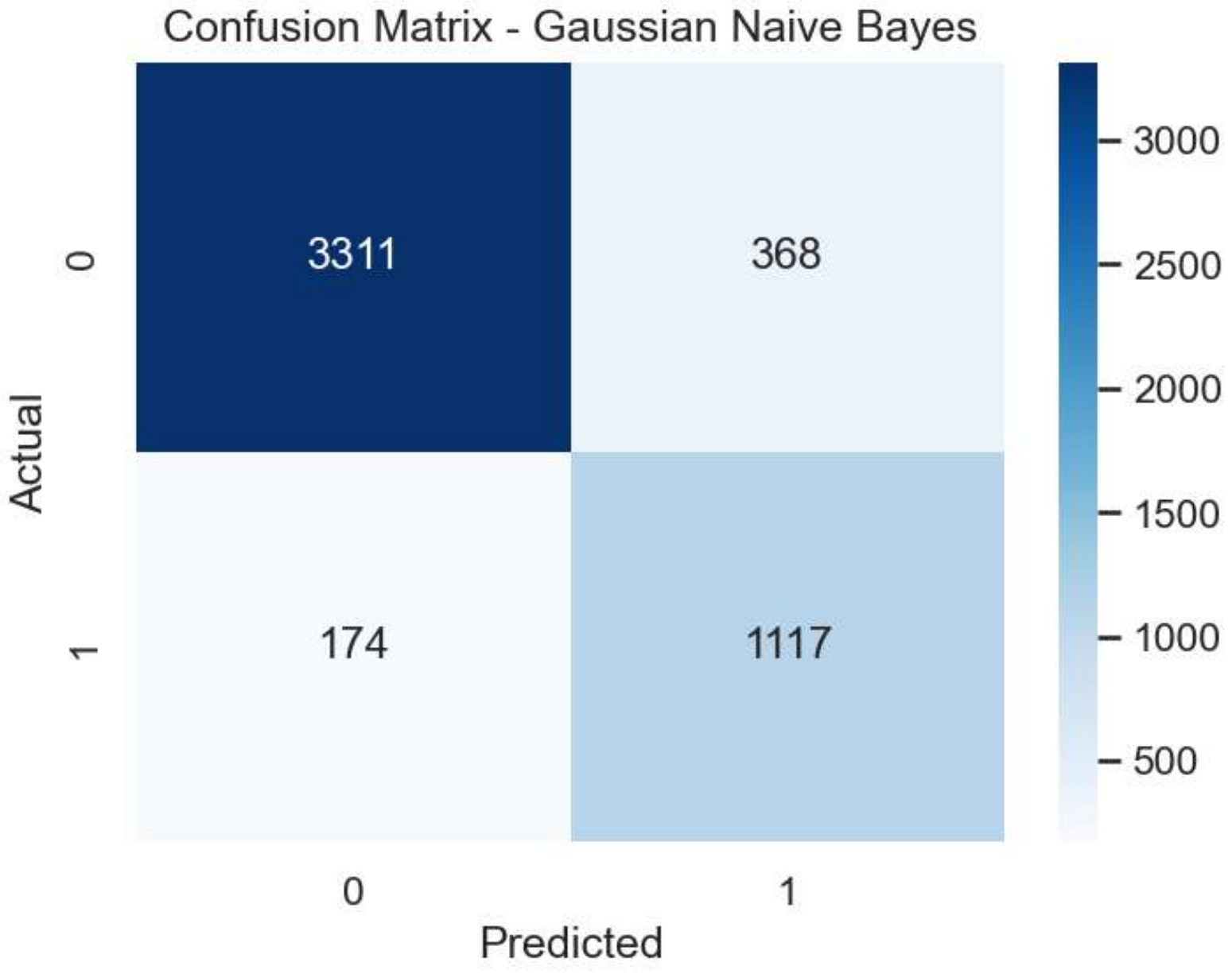

Average AQI by City
Ahmedabad
Delhi
Patna
Gurugram
Lucknow
Talcher
Jorapokhar
Brajrajnagar
Kolkata
Guwahati
Jaipur
Bhopal
Amritsar
Visakhapatnam
Chennai
Hyderabad
Mumbai
Kochi
Chandigarh
Amaravati
Bengaluru
Ernakulam
Thiruvananthapuram
Coimbatore
Shillong
Aizawl
City
0
100
200
300
400
Average AQI

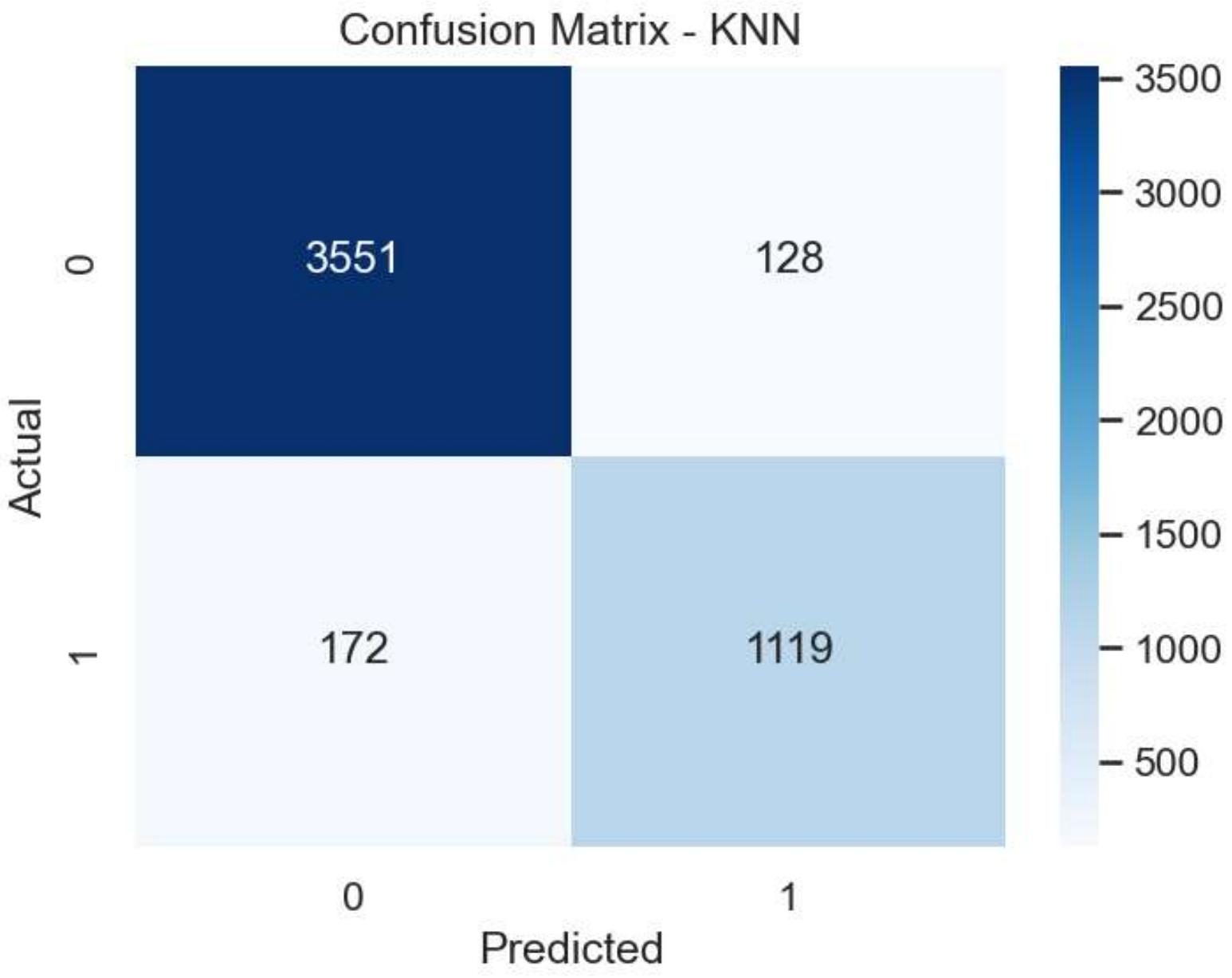

City-wise average AQI comparison
400
300
200
100
0
Aizawl
Shillong
Coimbatore
Thiruvananthapuram
Ernakulam
Bengaluru
Amaravati
Chandigarh
Kochi
Mumbai
Hyderabad
Chennai
Visakhapatnam
Amritsar
Bhopal
Jaipur
Guwahati
Kolkata
Brajrajnagar
Jorapokhar
Talcher
Lucknow
Gurugram
Patna
Delhi
Ahmedabad
City

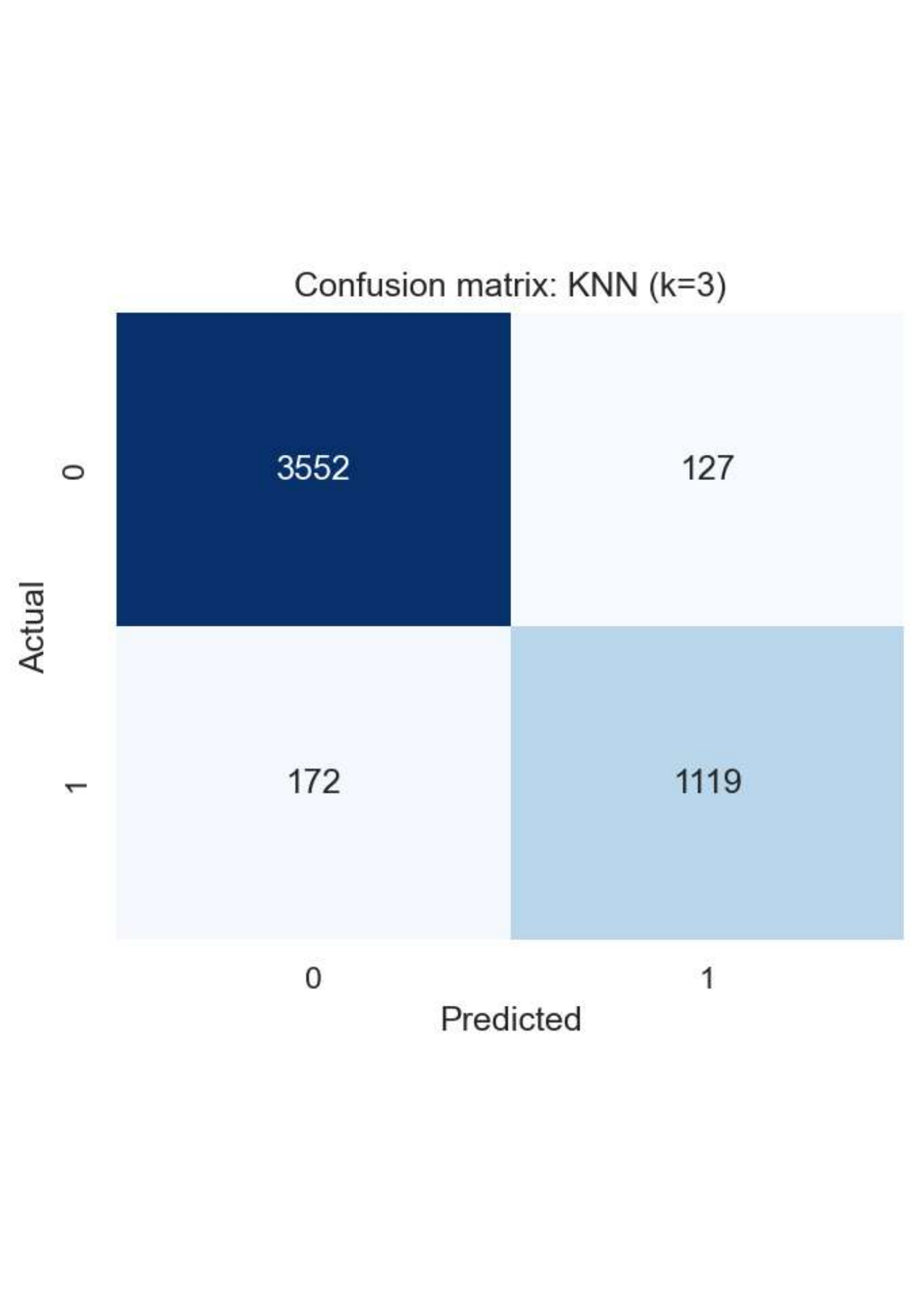

ROC curves: classification models
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC=0.976)
KNN (k=3) (AUC=0.959)
Gaussian Naive Bayes (AUC=0.958)
Decision Tree (AUC=0.977)
SVM (AUC=0.976)
MLPClassifier (64, 32) (AUC=0.985)

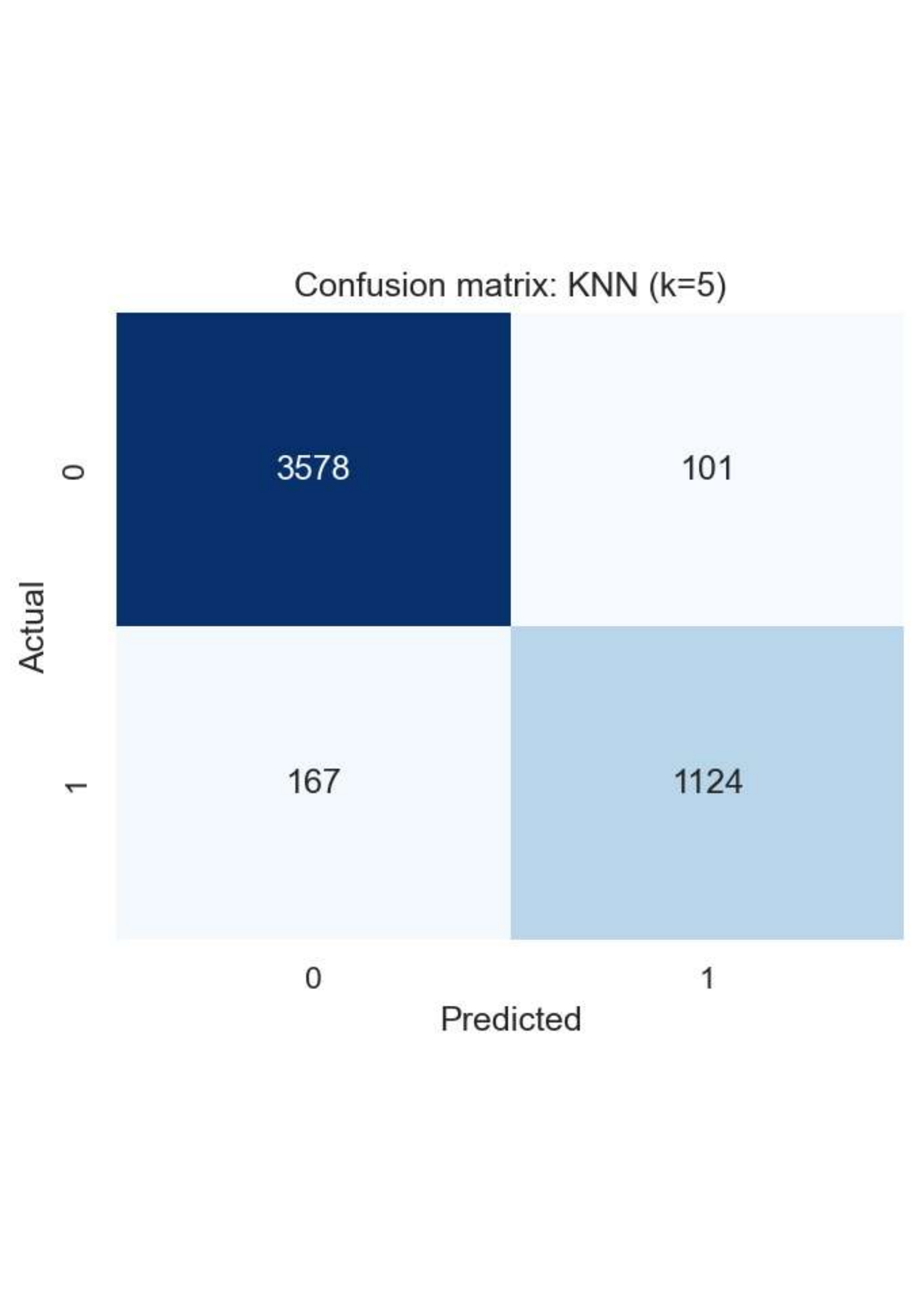

ROC Curves for Classification Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression AUC=0.976
KNN AUC=0.96
Gaussian Naive Bayes AUC=0.958
Decision Tree AUC=0.977
SVM AUC=0.976
MLPClassifier AUC=0.985

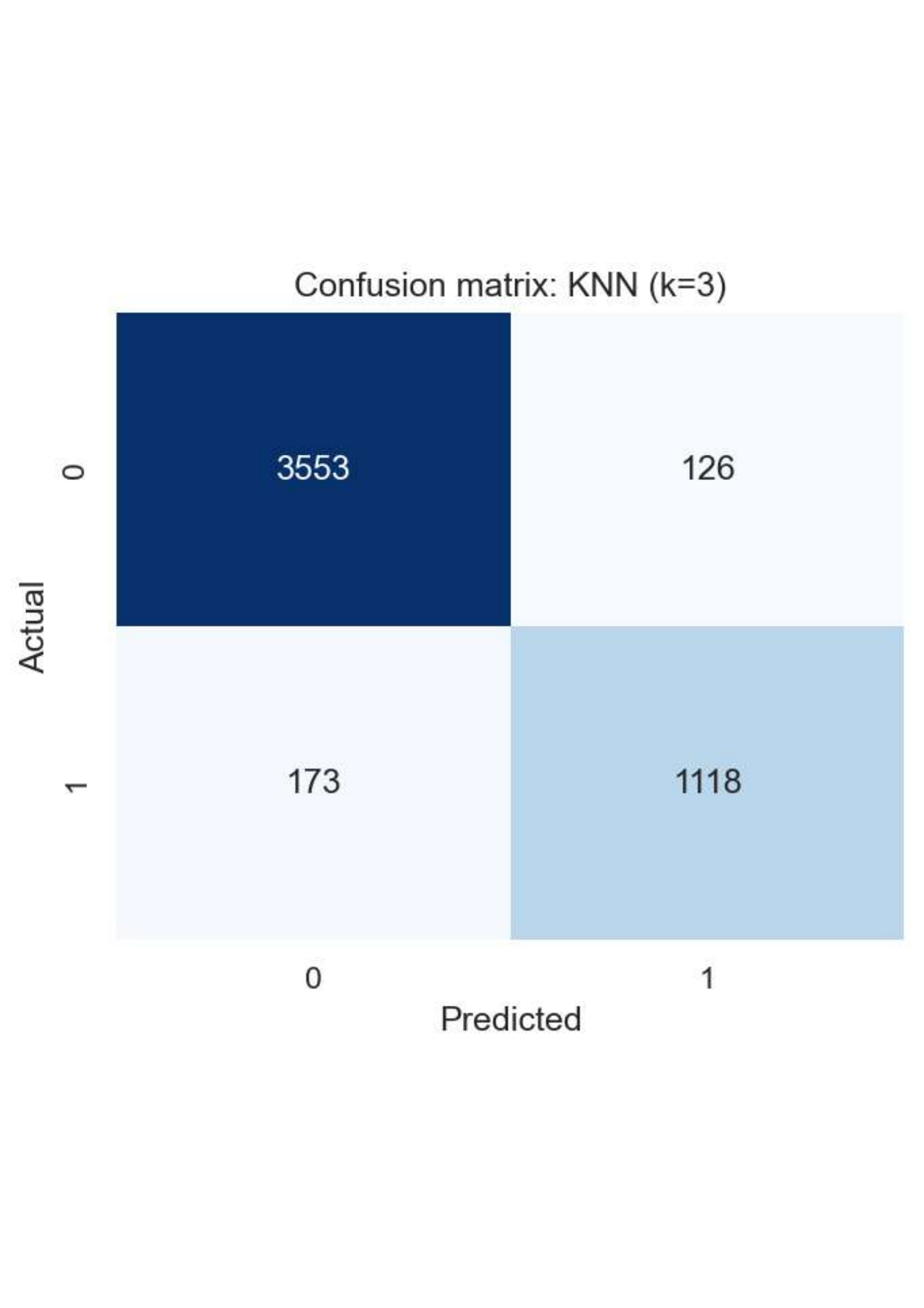

Confusion Matrix - Decision Tree
3508
171
116
1175
0
1
Actual
0
1
Predicted
3500
3000
2500
2000
1500
1000
500

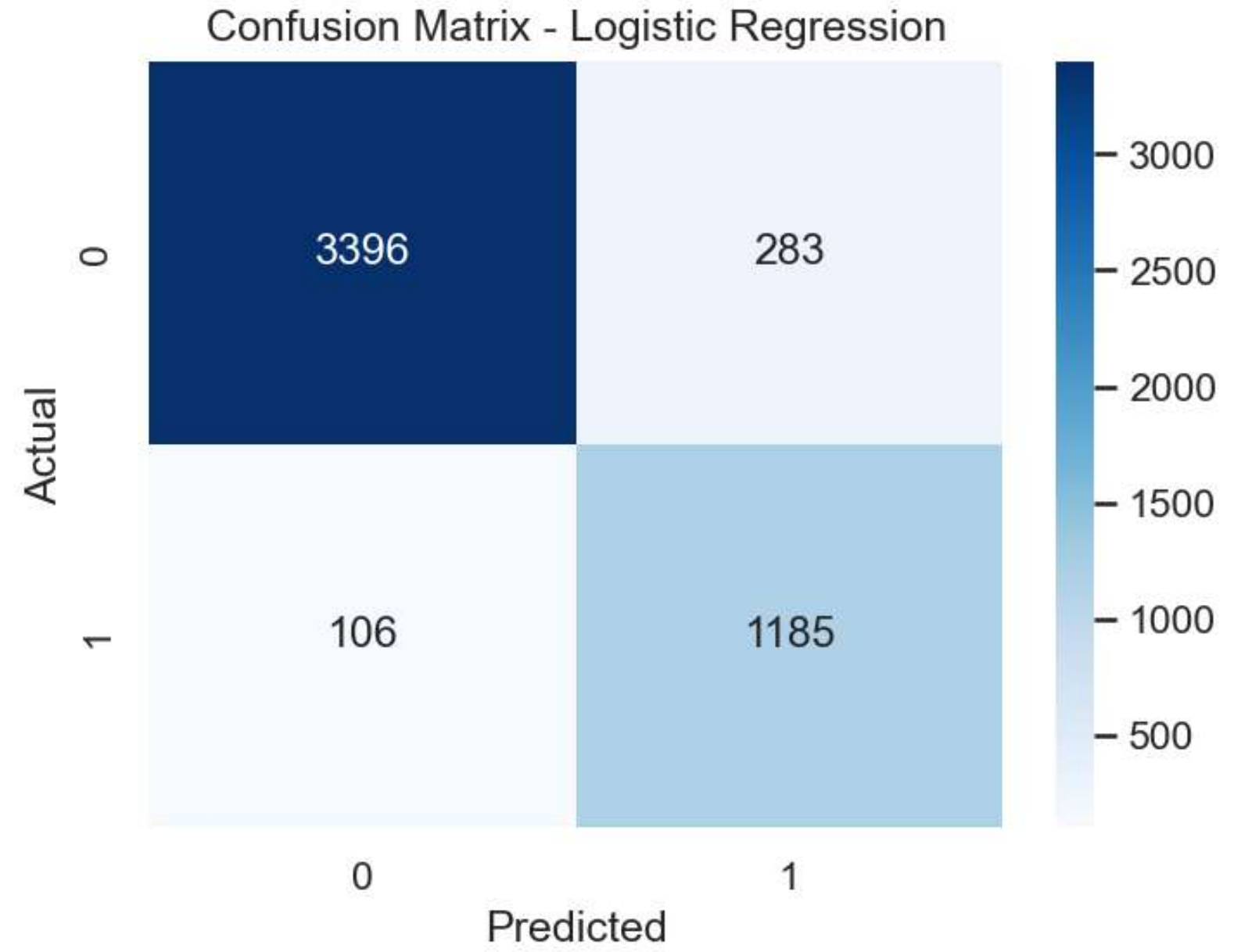

Confusion Matrix - Gaussian Naive Bayes
3311
368
174
1117
0
1
Actual
0
1
Predicted
3000
2500
2000
1500
1000
500

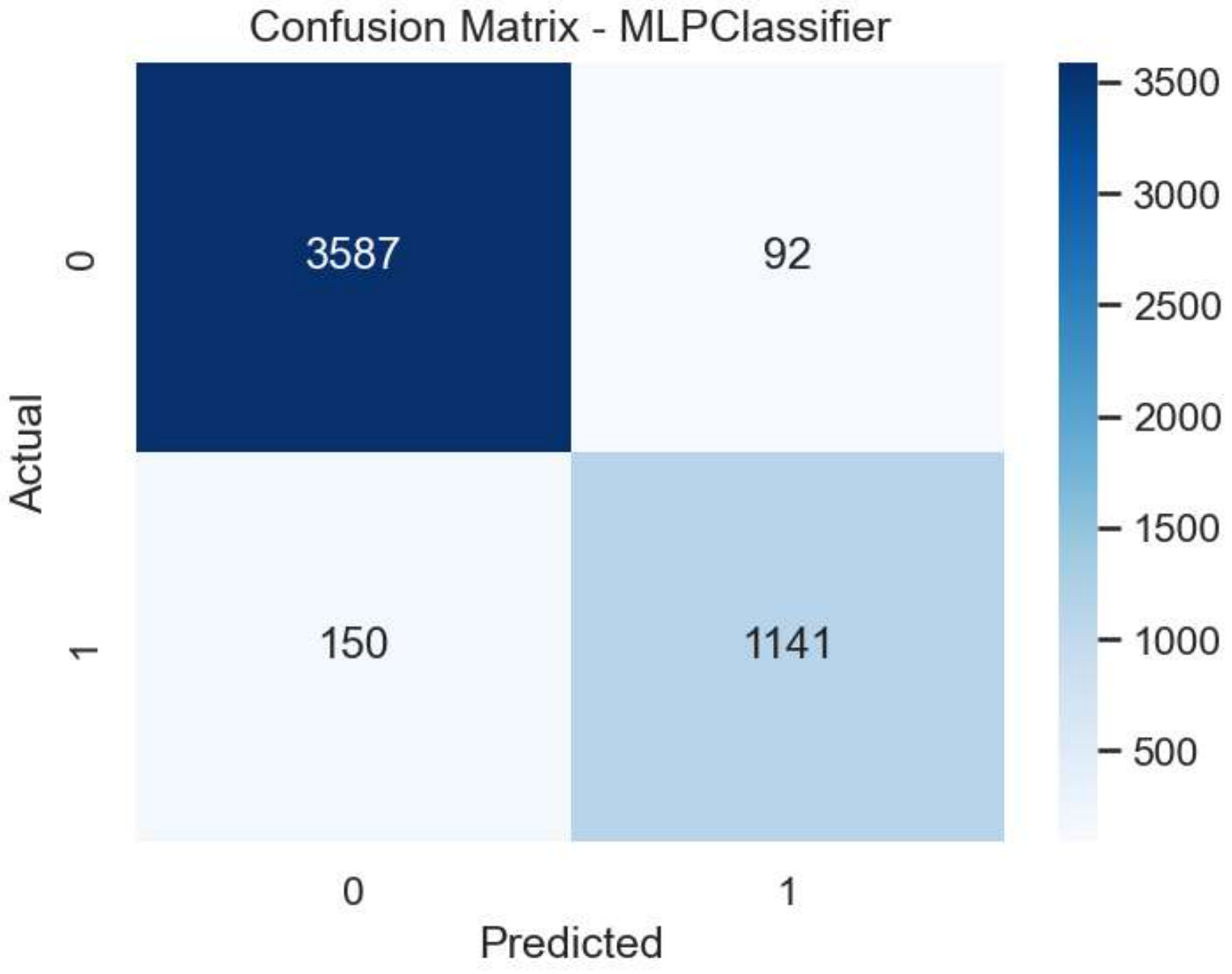

Confusion Matrix - KNN
3551
128
172
1119
0
1
Actual
0
1
Predicted
3500
3000
2500
2000
1500
1000
500

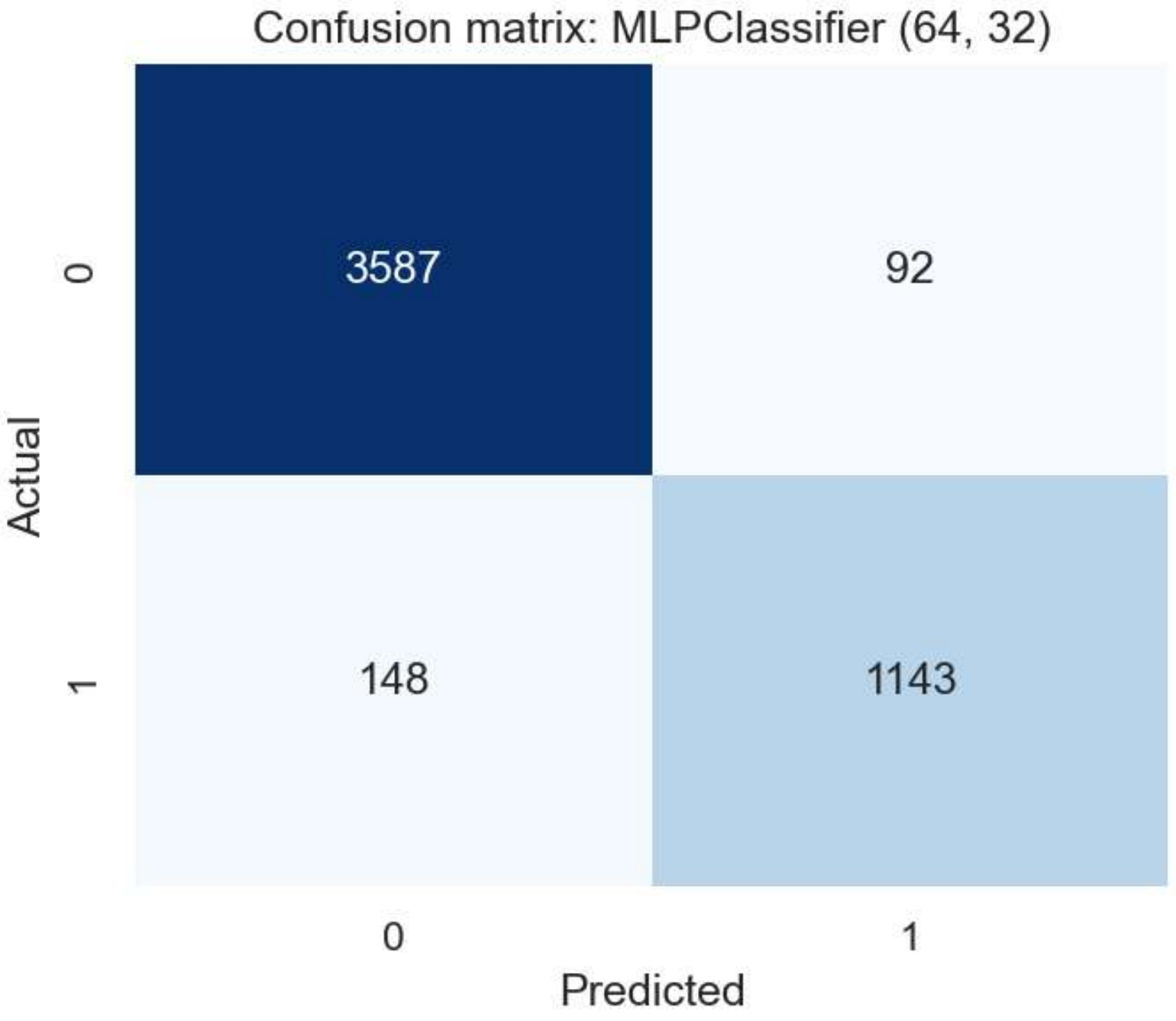

Confusion matrix: KNN (k=3)
3552
127
172
1119
0
1
Actual
0
1
Predicted

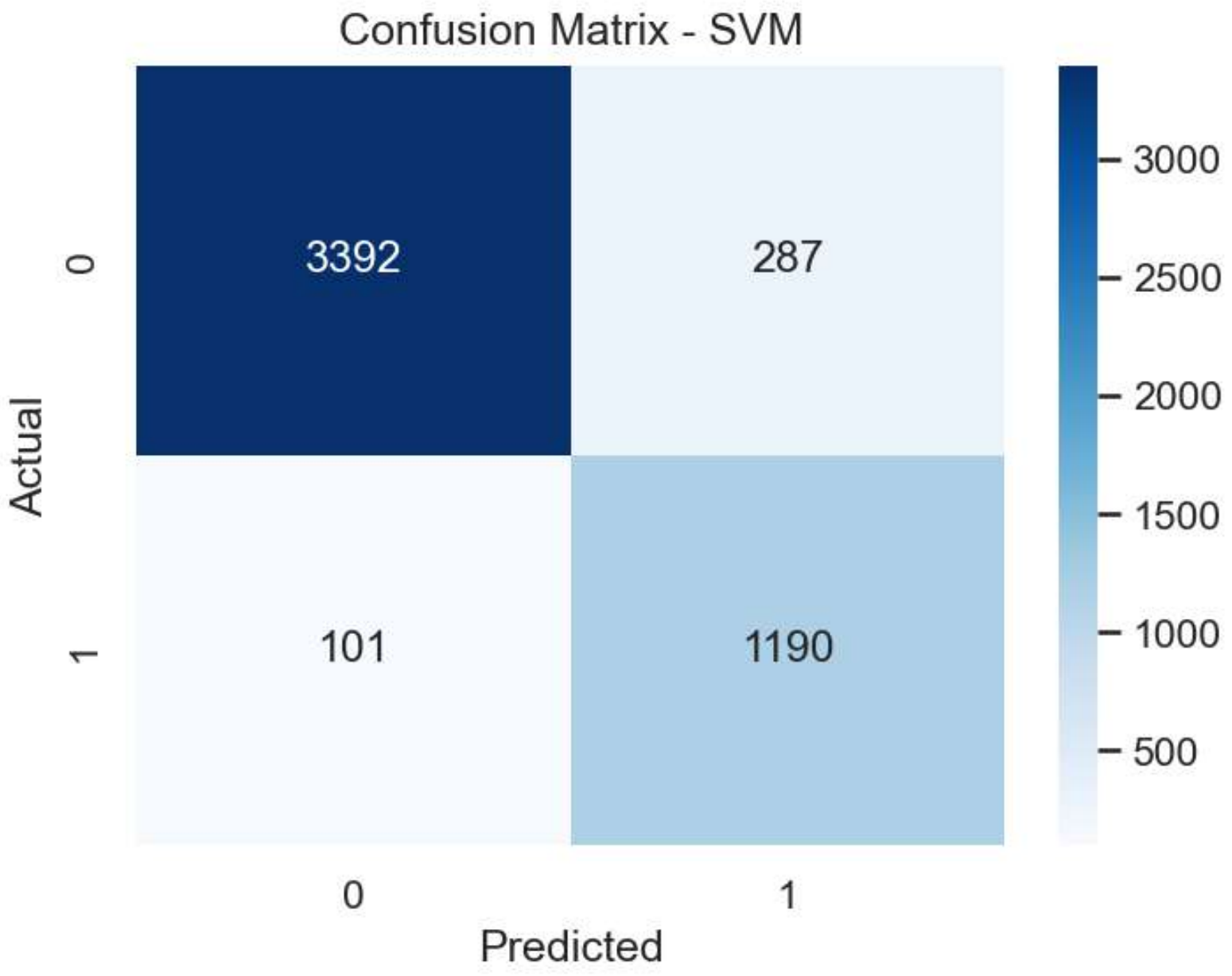

Confusion matrix: KNN (k=5)
3578
101
167
1124
Actual
0
1
0
1
Predicted

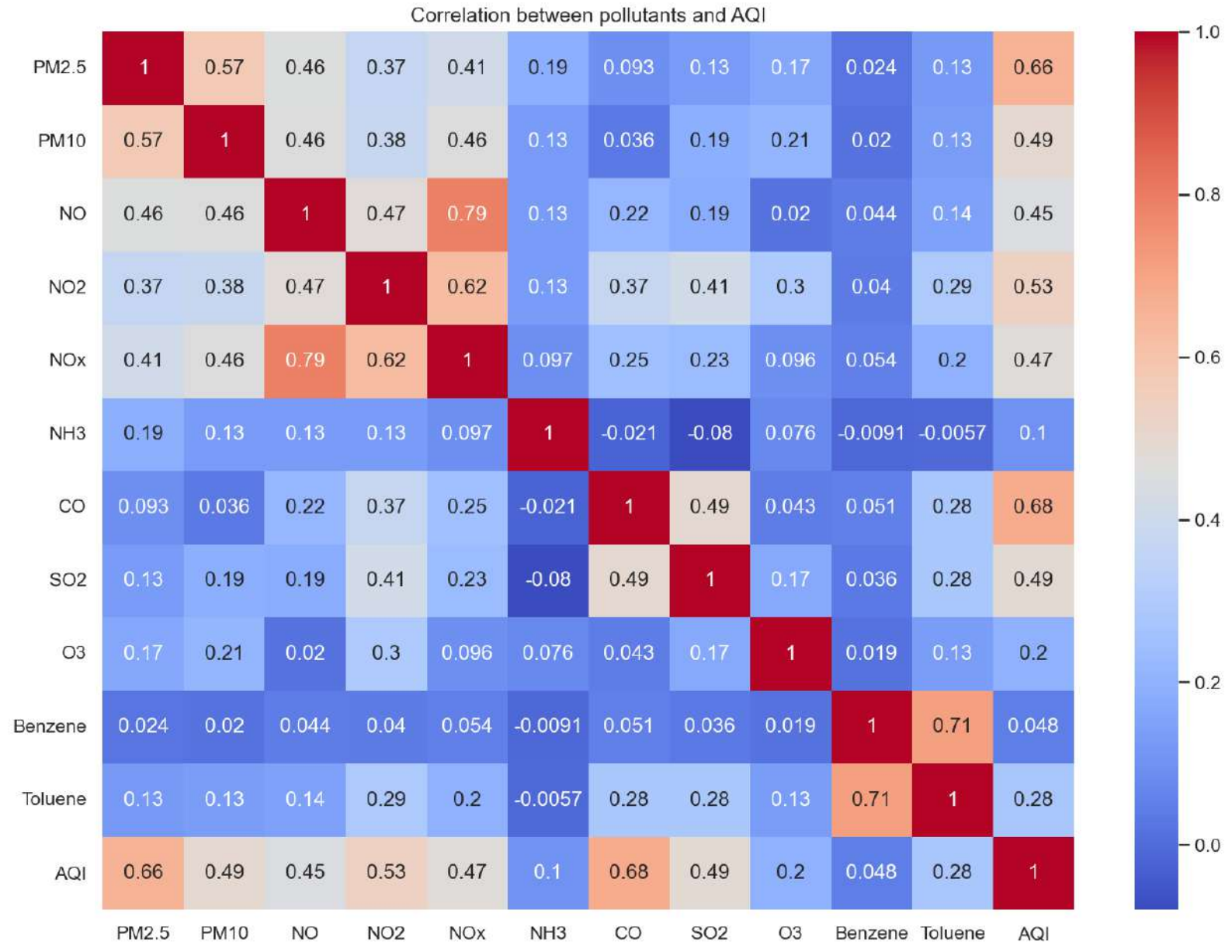

Confusion matrix: KNN (k=3)
3553
126
173
1118
0
1
Actual
0
1
Predicted

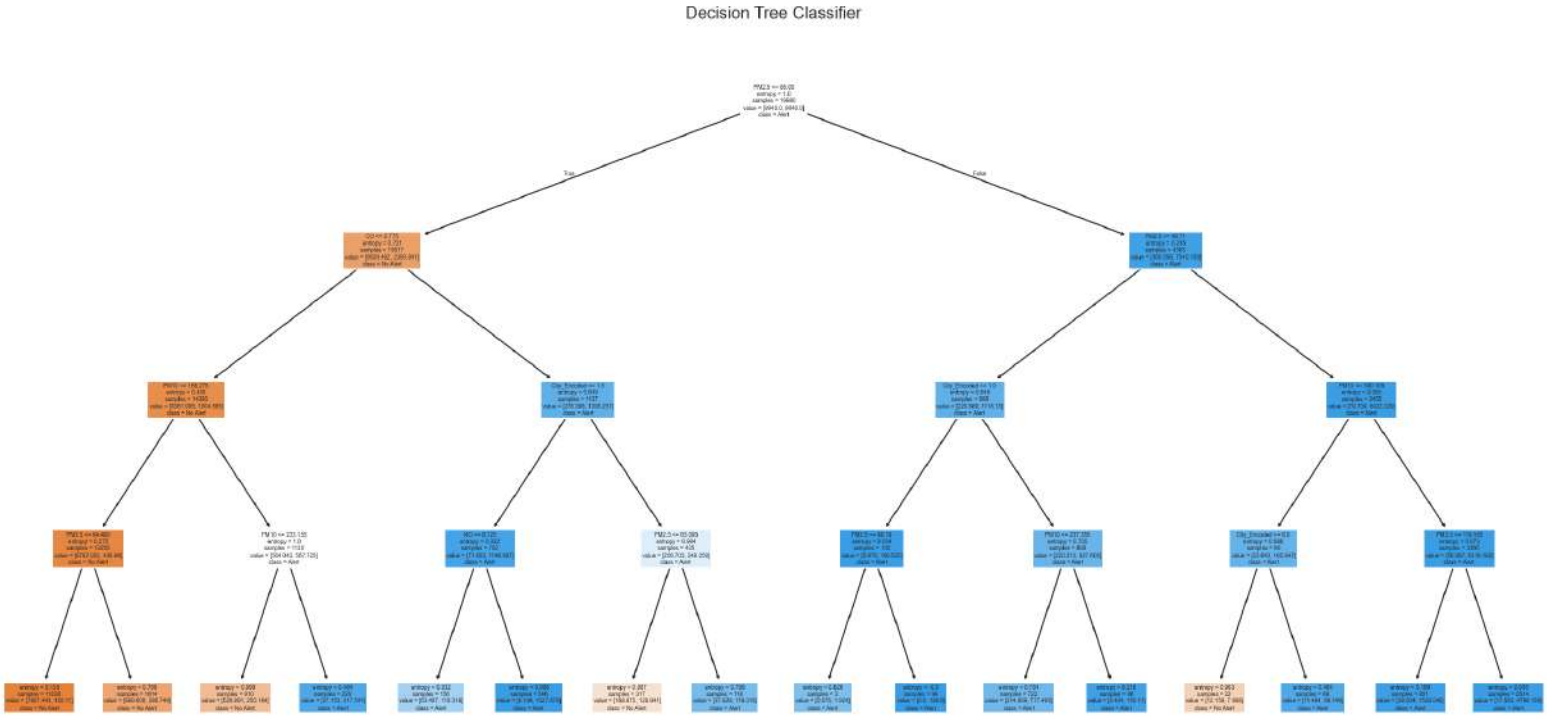

Confusion Matrix - Logistic Regression
3396
283
106
1185
0
1
Actual
0
1
Predicted
3000
2500
2000
1500
1000
500

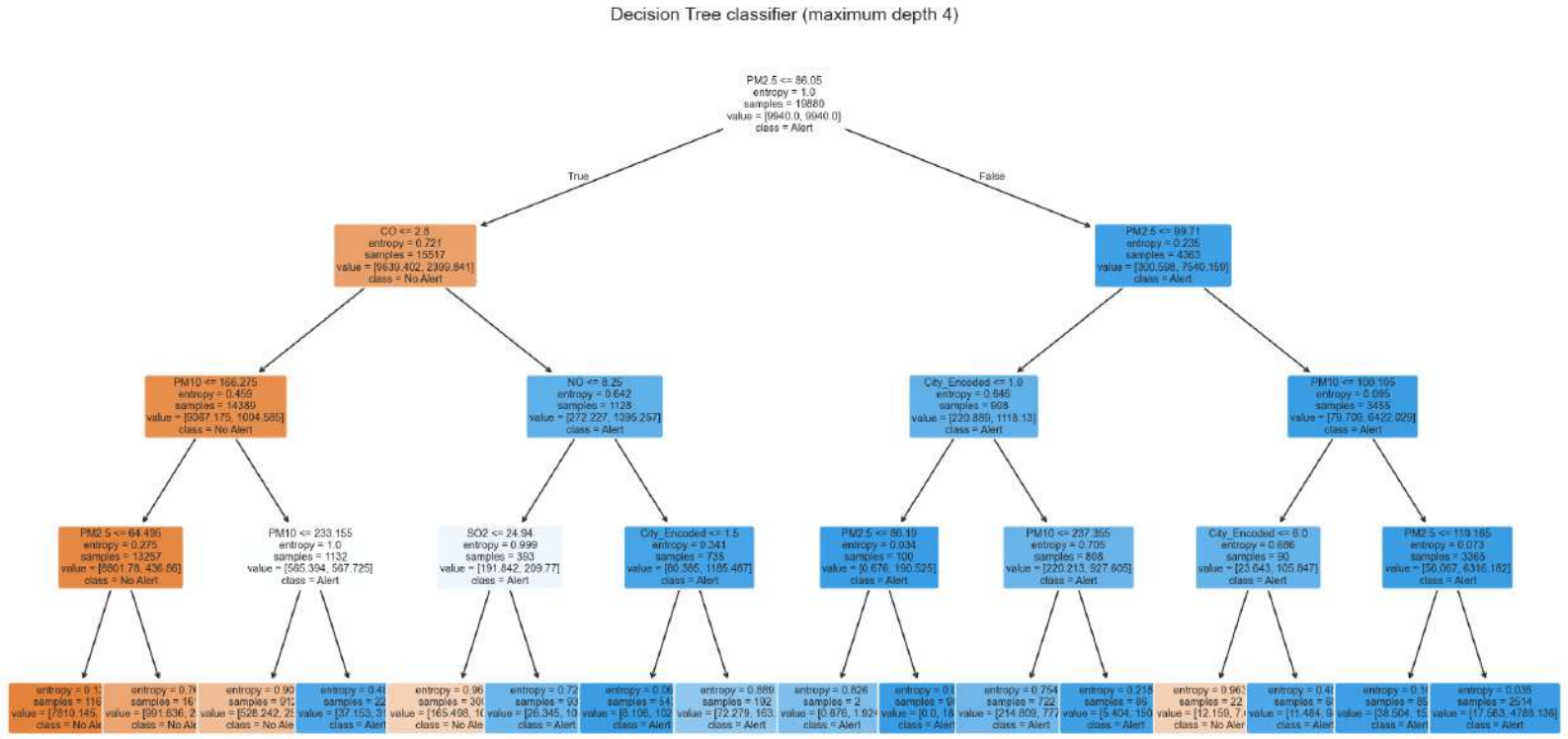

Confusion Matrix - MLPClassifier
3587
92
150
1141
0
1
Actual
0
1
Predicted
3500
3000
2500
2000
1500
1000
500

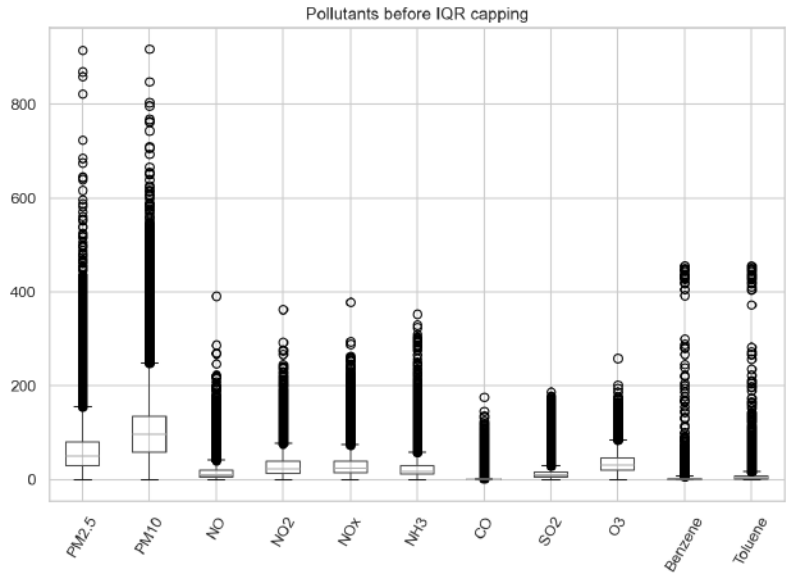

Confusion matrix: MLPClassifier (64, 32)
Actual
0
1
3587
92
148
1143
0
1
Predicted

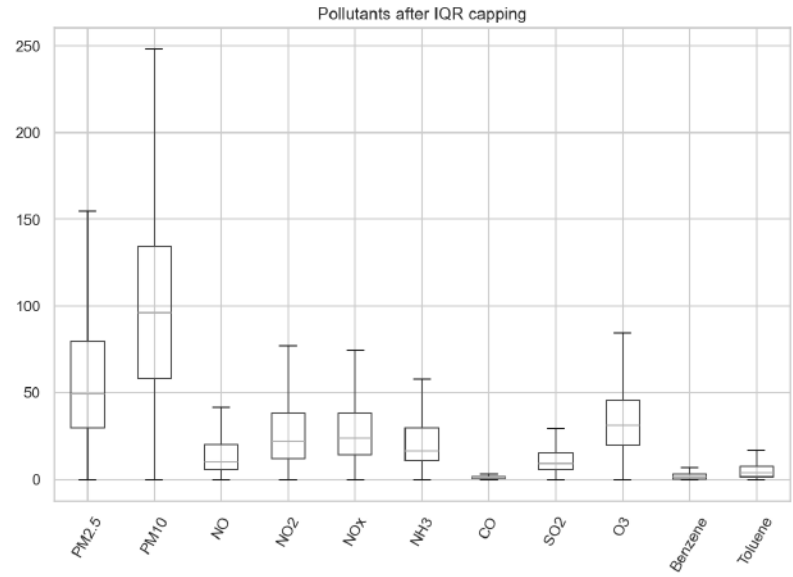

Confusion Matrix - SVM
3392
287
101
1190
0
1
Actual
0
1
Predicted
3000
2500
2000
1500
1000
500

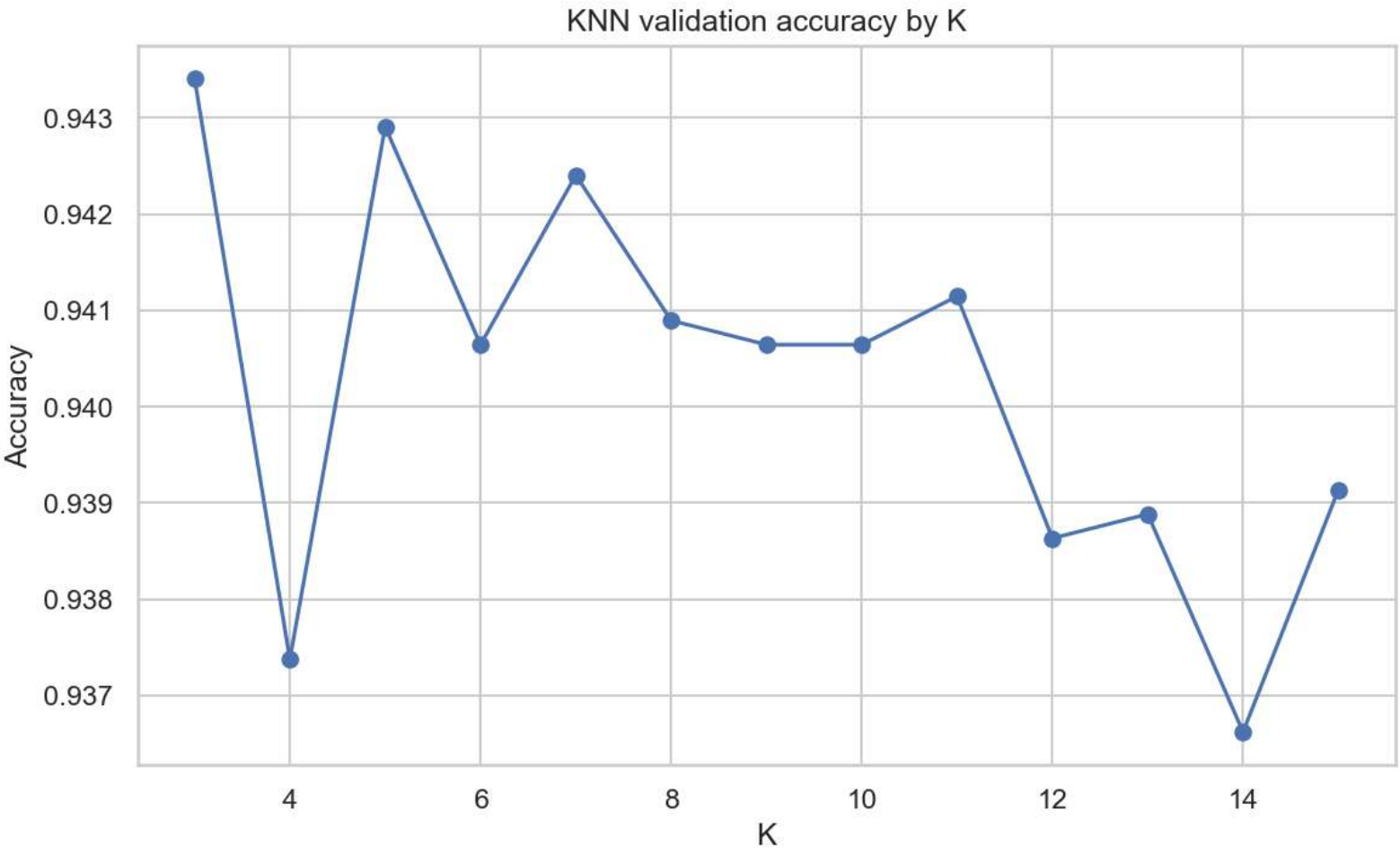

Correlation between pollutants and AQI
PM2.5 1 0.57 0.46 0.37 0.41 0.19 0.093 0.13 0.17 0.024 0.13 0.66
PM10 0.57 1 0.46 0.38 0.46 0.13 0.036 0.19 0.21 0.02 0.13 0.49
NO 0.46 0.46 1 0.47 0.79 0.13 0.22 0.19 0.02 0.044 0.14 0.45
NO2 0.37 0.38 0.47 1 0.62 0.13 0.37 0.41 0.3 0.04 0.29 0.53
NOx 0.41 0.46 0.79 0.62 1 0.097 0.25 0.23 0.096 0.054 0.2 0.47
NH3 0.19 0.13 0.13 0.13 0.097 1 -0.021 -0.08 0.076 -0.0091 -0.0057 0.1
CO 0.093 0.036 0.22 0.37 0.25 -0.021 1 0.49 0.043 0.051 0.28 0.68
SO2 0.13 0.19 0.19 0.41 0.23 -0.08 0.49 1 0.17 0.036 0.28 0.49
O3 0.17 0.21 0.02 0.3 0.096 0.076 0.043 0.17 1 0.019 0.13 0.2
Benzene 0.024 0.02 0.044 0.04 0.054 -0.0091 0.051 0.036 0.019 1 0.71 0.048
Toluene 0.13 0.13 0.14 0.29 0.2 -0.0057 0.28 0.28 0.13 0.71 1 0.28
AQI 0.66 0.49 0.45 0.53 0.47 0.1 0.68 0.49 0.2 0.048 0.28 1
PM2.5 PM10 NO NO2 NOx NH3 CO SO2 O3 Benzene Toluene AQI
1.0
0.8
0.6
0.4
0.2
0.0

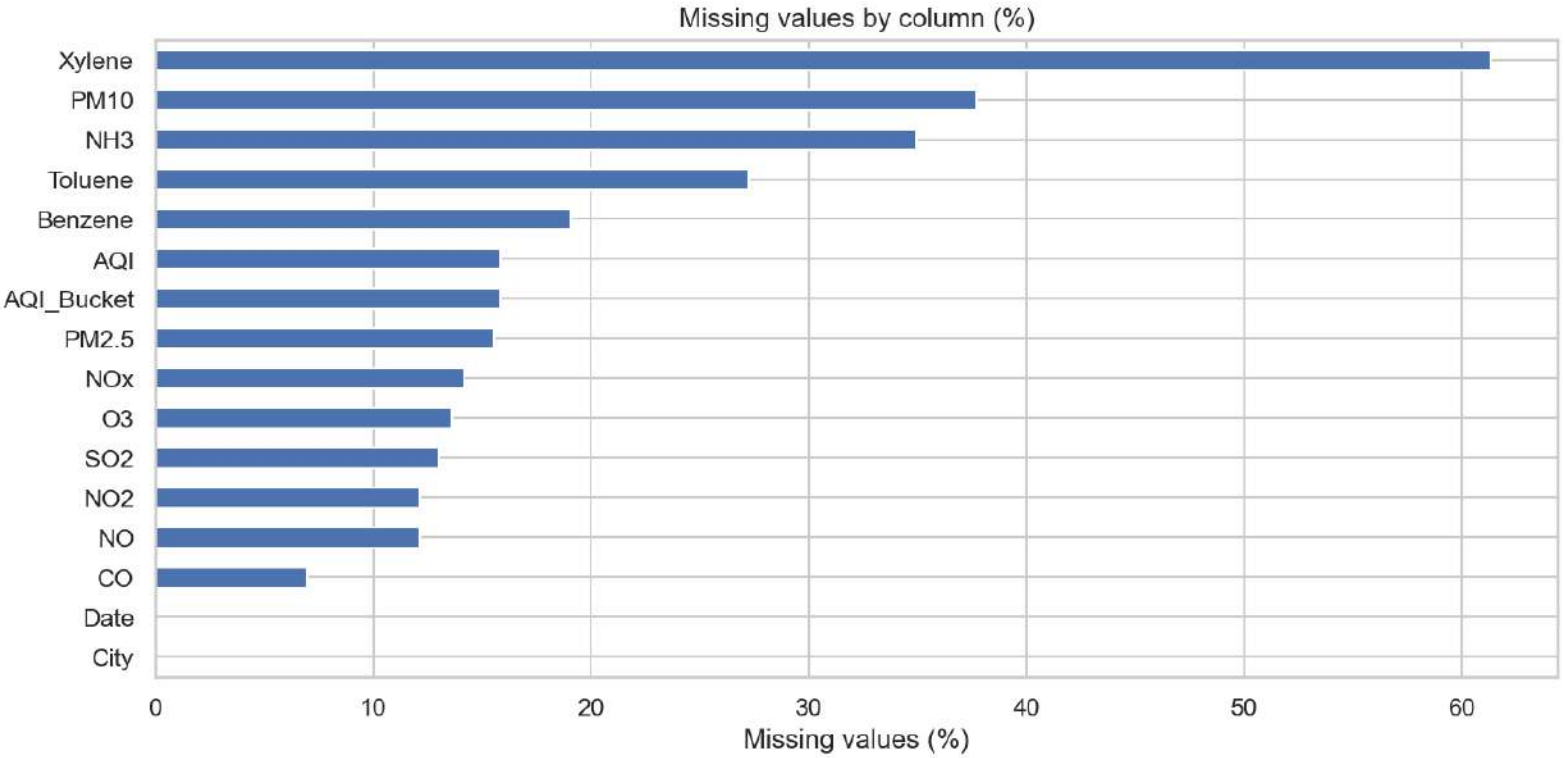

Decision Tree Classifier

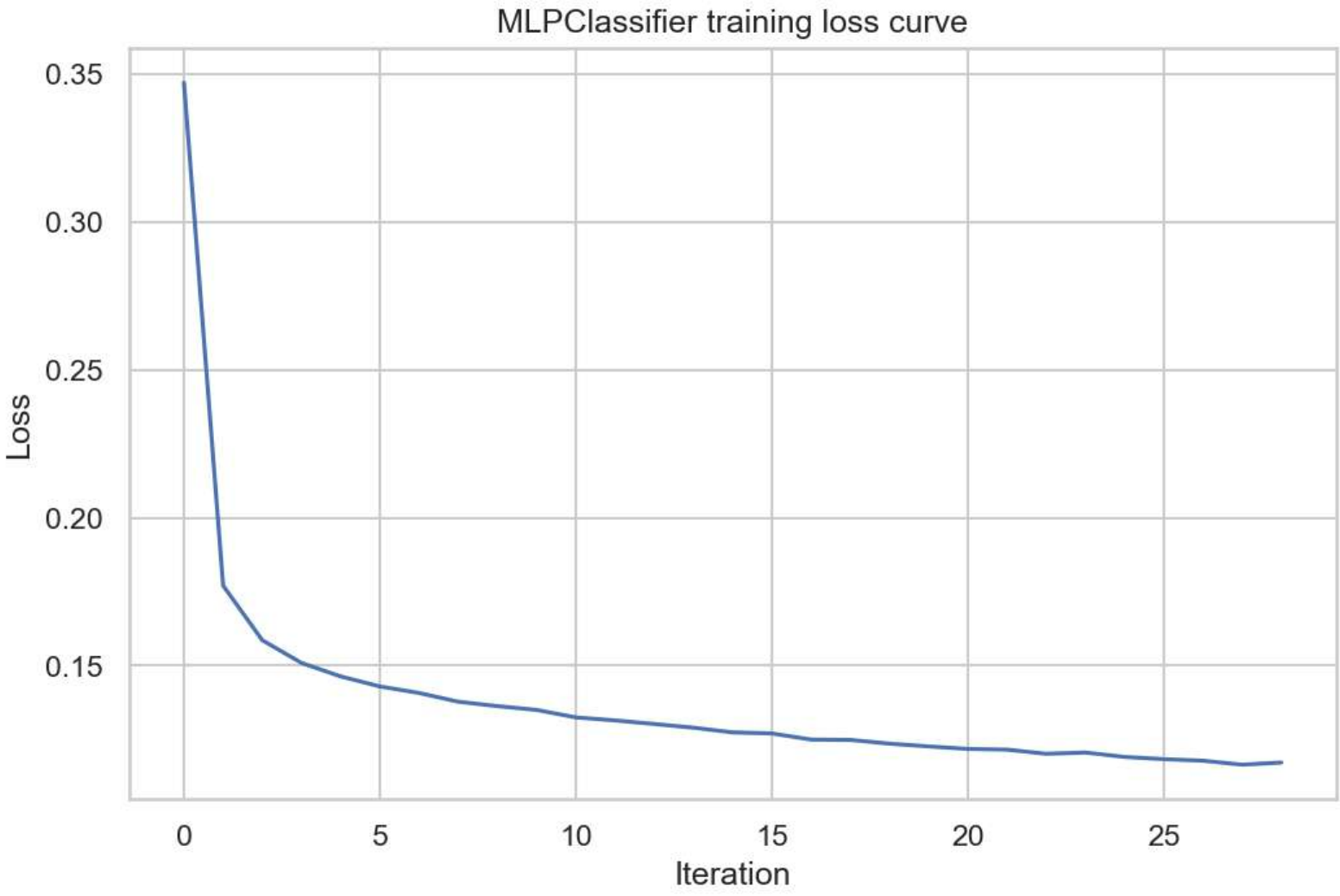

Decision Tree classifier (maximum depth 4)
PM2.5 <= 86.05
entropy = 1.0
samples = 19880
value = [9940.0, 9940.0]
class = Alert
True
False
CO <= 2.8
entropy = 0.721
samples = 15517
value = [9639.402, 2309.841]
class = No Alert
PM2.5 <= 99.71
entropy = 0.235
samples = 4363
value = [300.598, 7540.159]
class = Alert
PM10 <= 166.275
entropy = 0.459
samples = 14389
value = [9367.175, 1004.585]
class = No Alert
NO <= 8.25
entropy = 0.642
samples = 1128
value = [272.227, 1395.257]
class = Alert
City_Encoded <= 1.0
entropy = 0.845
samples = 908
value = [220.889, 1118.13]
class = Alert
PM10 <= 100.165
entropy = 0.095
samples = 3455
value = [79.709, 6422.029]
class = Alert
PM2.5 <= 64.495
entropy = 0.275
samples = 13257
value = [8801.78, 436.86]
class = No Alert
PM10 <= 233.155
entropy = 1.0
samples = 1132
value = [565.394, 567.725]
class = Alert
SO2 <= 24.94
entropy = 0.999
samples = 393
value = [191.842, 209.77]
class = Alert
City_Encoded <= 1.5
entropy = 0.341
samples = 735
value = [80.385, 1185.487]
class = Alert
PM2.5 <= 86.19
entropy = 0.034
samples = 100
value = [0.676, 190.525]
class = Alert
PM10 <= 237.355
entropy = 0.705
samples = 808
value = [220.213, 927.605]
class = Alert
City_Encoded <= 6.0
entropy = 0.686
samples = 90
value = [23.643, 105.847]
class = Alert
PM2.5 <= 119.165
entropy = 0.073
samples = 3365
value = [56.067, 6316.182]
class = Alert

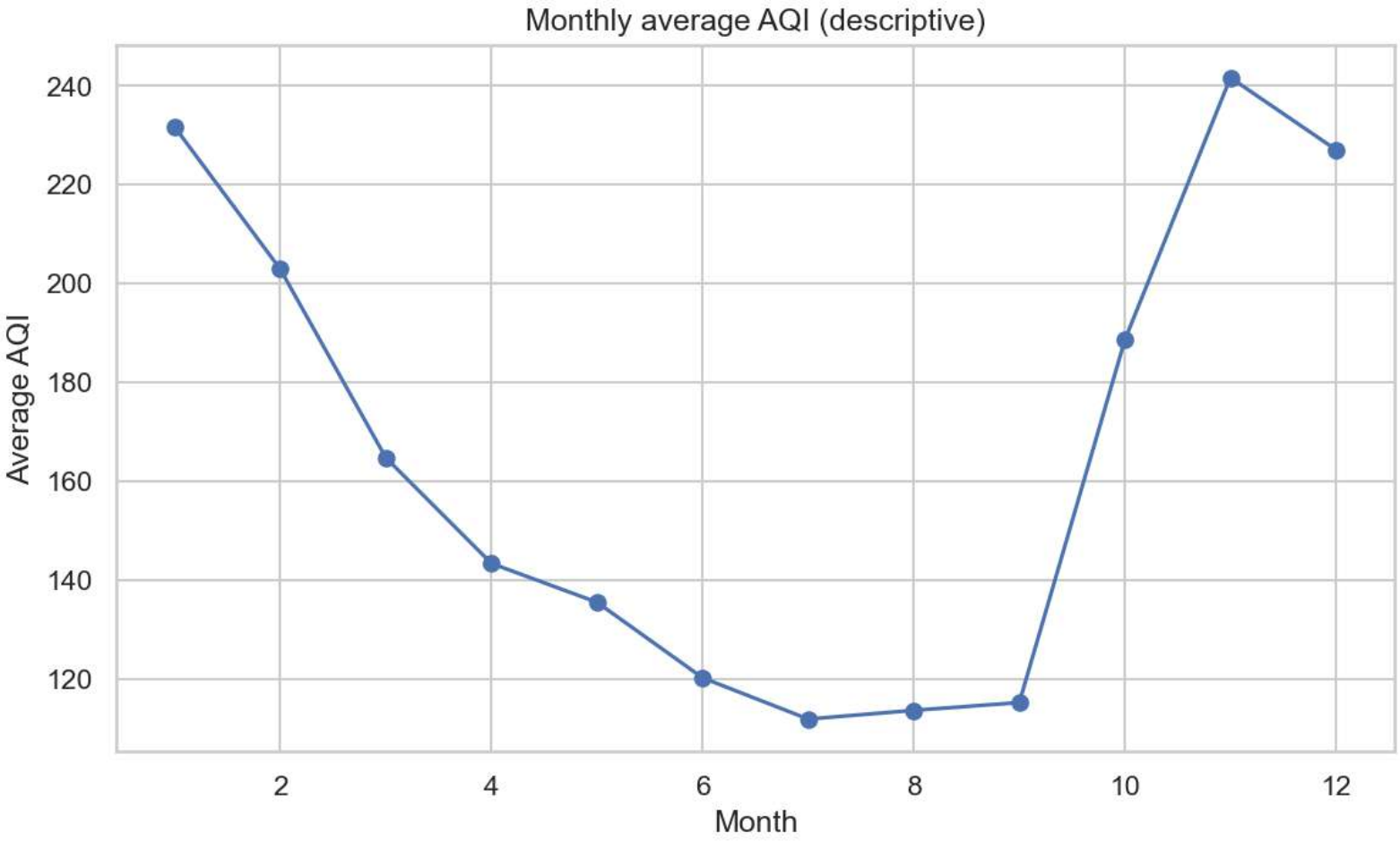

Pollutants before IQR capping
800
600
400
200
0
PM2.5
PM10
NO
NO2
NOx
NH3
CO
SO2
O3
Benzene
Toluene

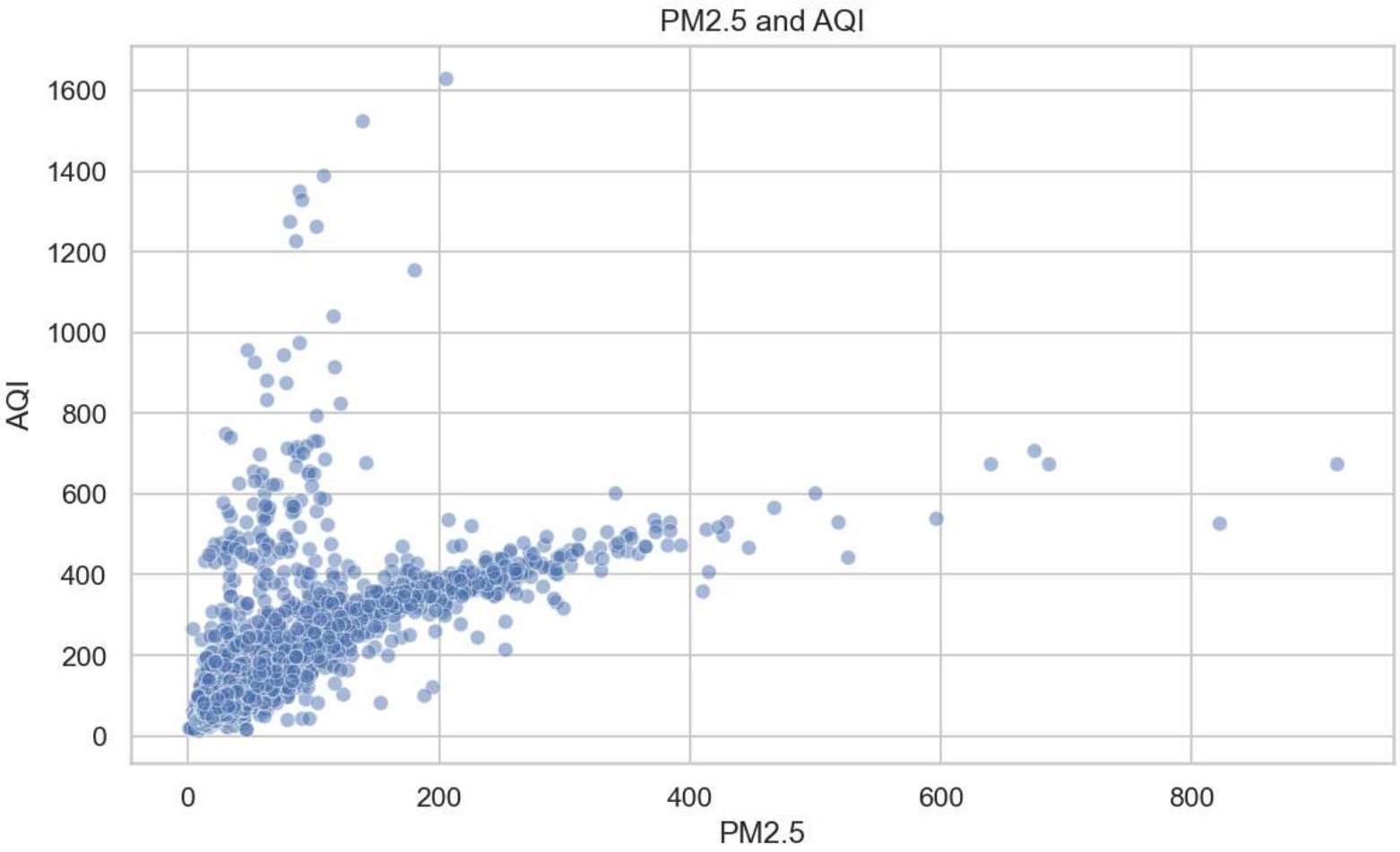

Pollutants after IQR capping
250
200
150
100
50
0
PM2.5
PM10
NO
NO2
NOx
NH3
CO
SO2
O3
Benzene
Toluene

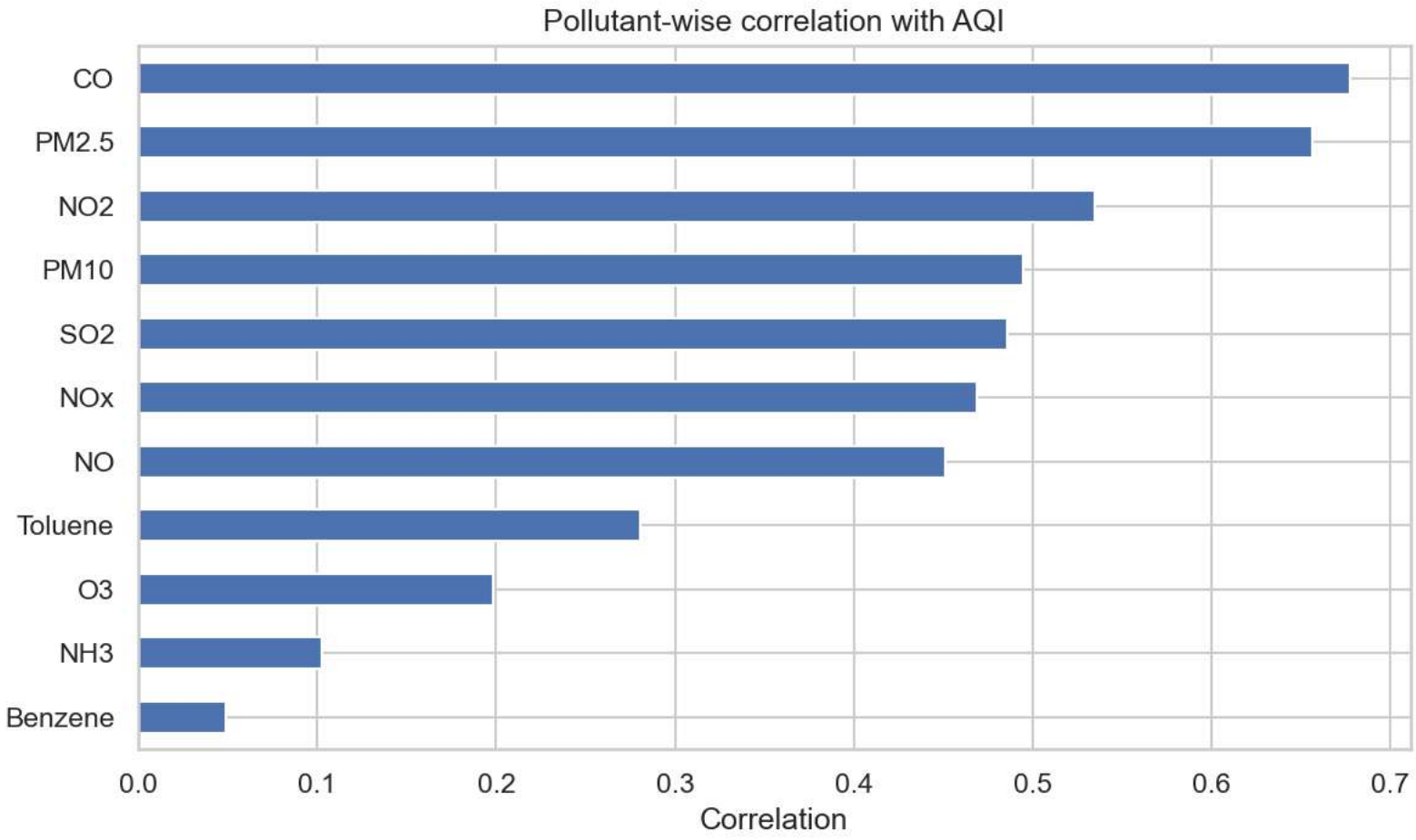

KNN validation accuracy by K
Accuracy
0.943
0.942
0.941
0.940
0.939
0.938
0.937
4
6
8
10
12
14
K

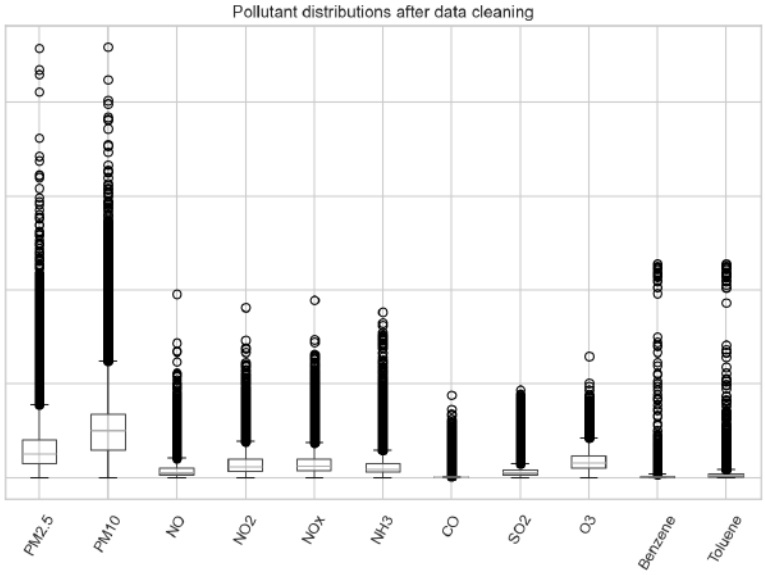

Missing values by column (%)
Xylene
PM10
NH3
Toluene
Benzene
AQI
AQI_Bucket
PM2.5
NOx
O3
SO2
NO2
NO
CO
Date
City
0
10
20
30
40
50
60
Missing values (%)


MLPClassifier training loss curve
0.35
0.30
0.25
0.20
0.15
Loss
0
5
10
15
20
25
Iteration

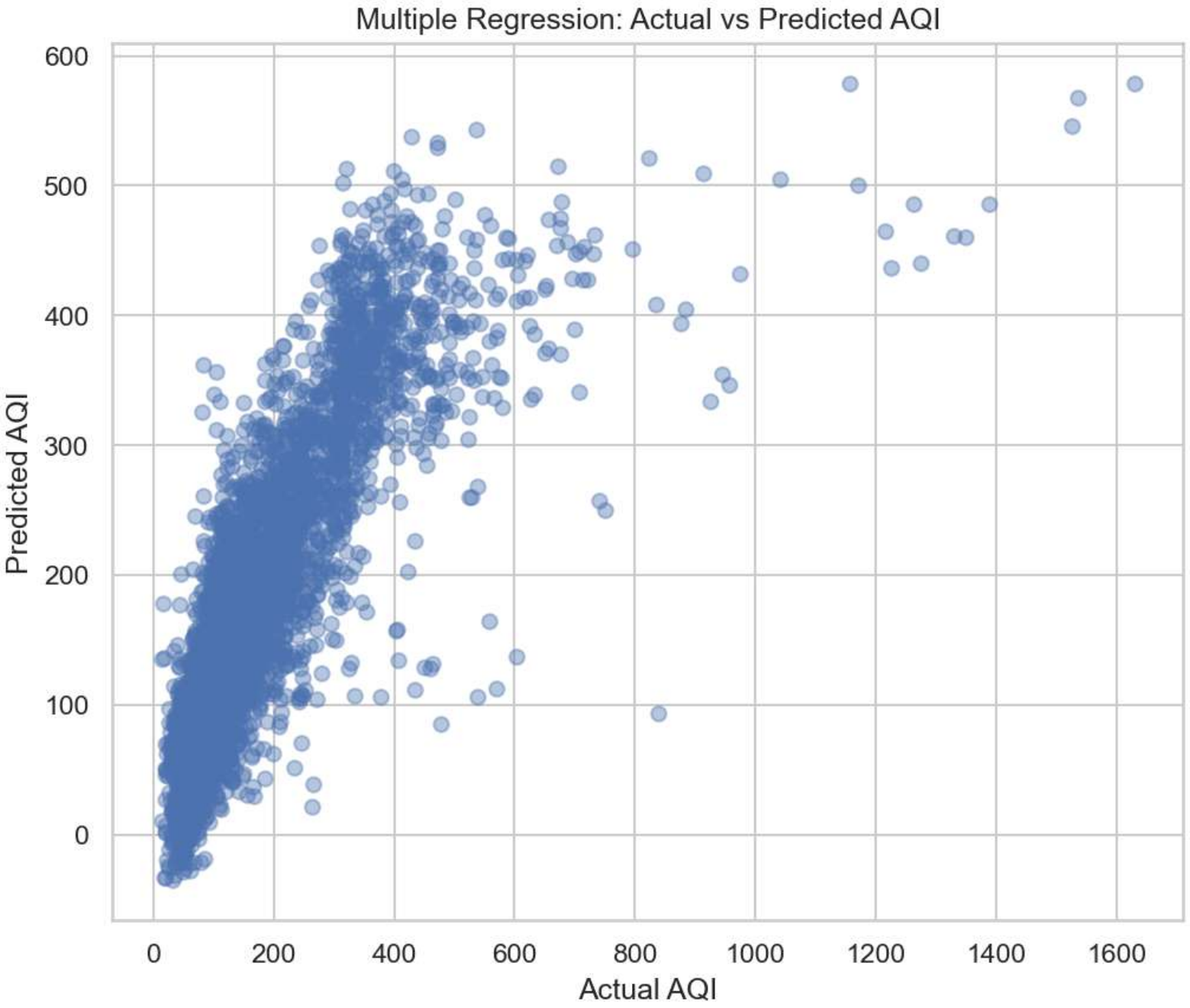

Monthly average AQI (descriptive)
Average AQI
240
220
200
180
160
140
120
2
4
6
8
10
12
Month

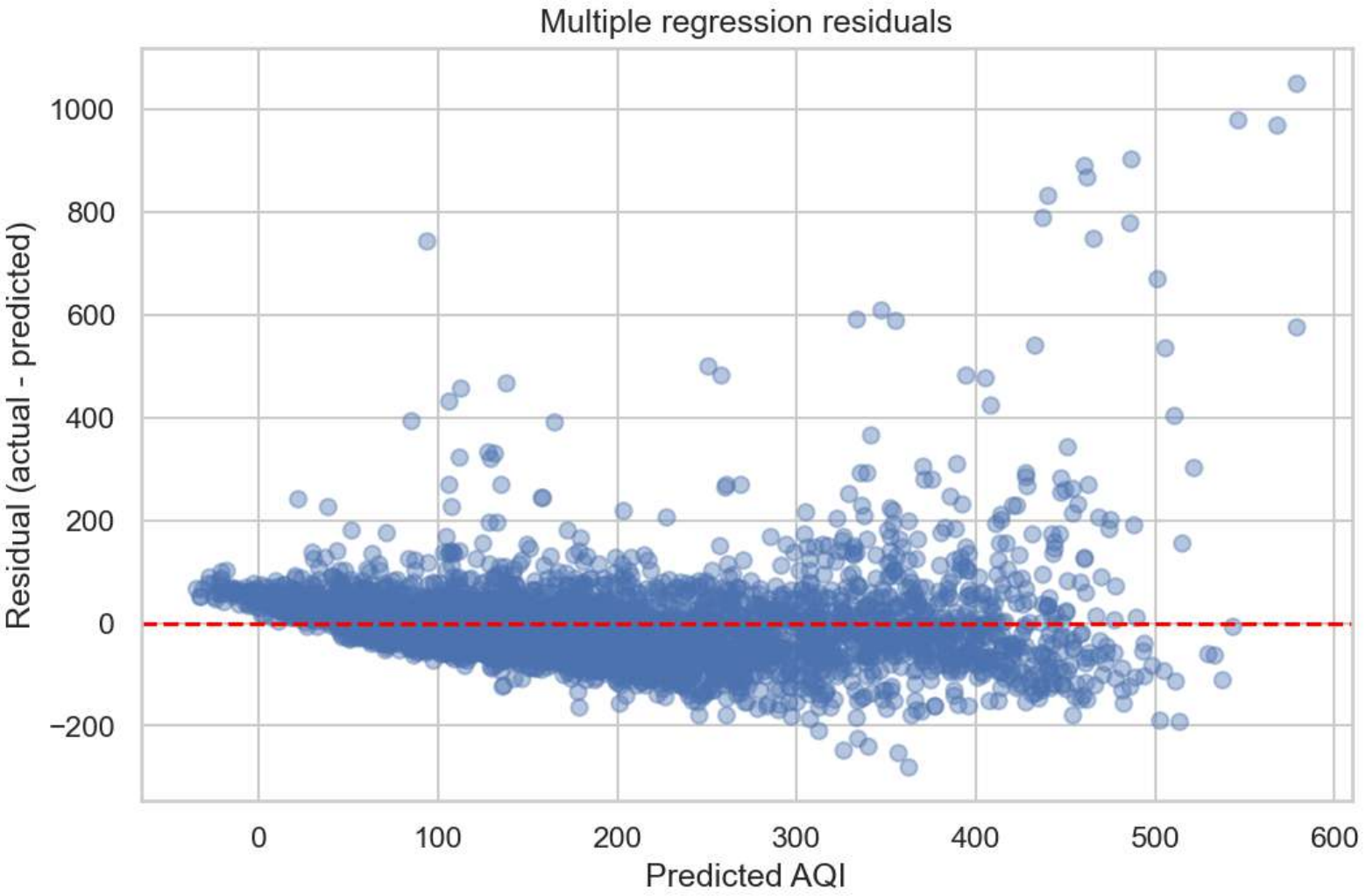

PM2.5 and AQI
AQI
PM2.5
0
200
400
600
800
1000
1200
1400
1600

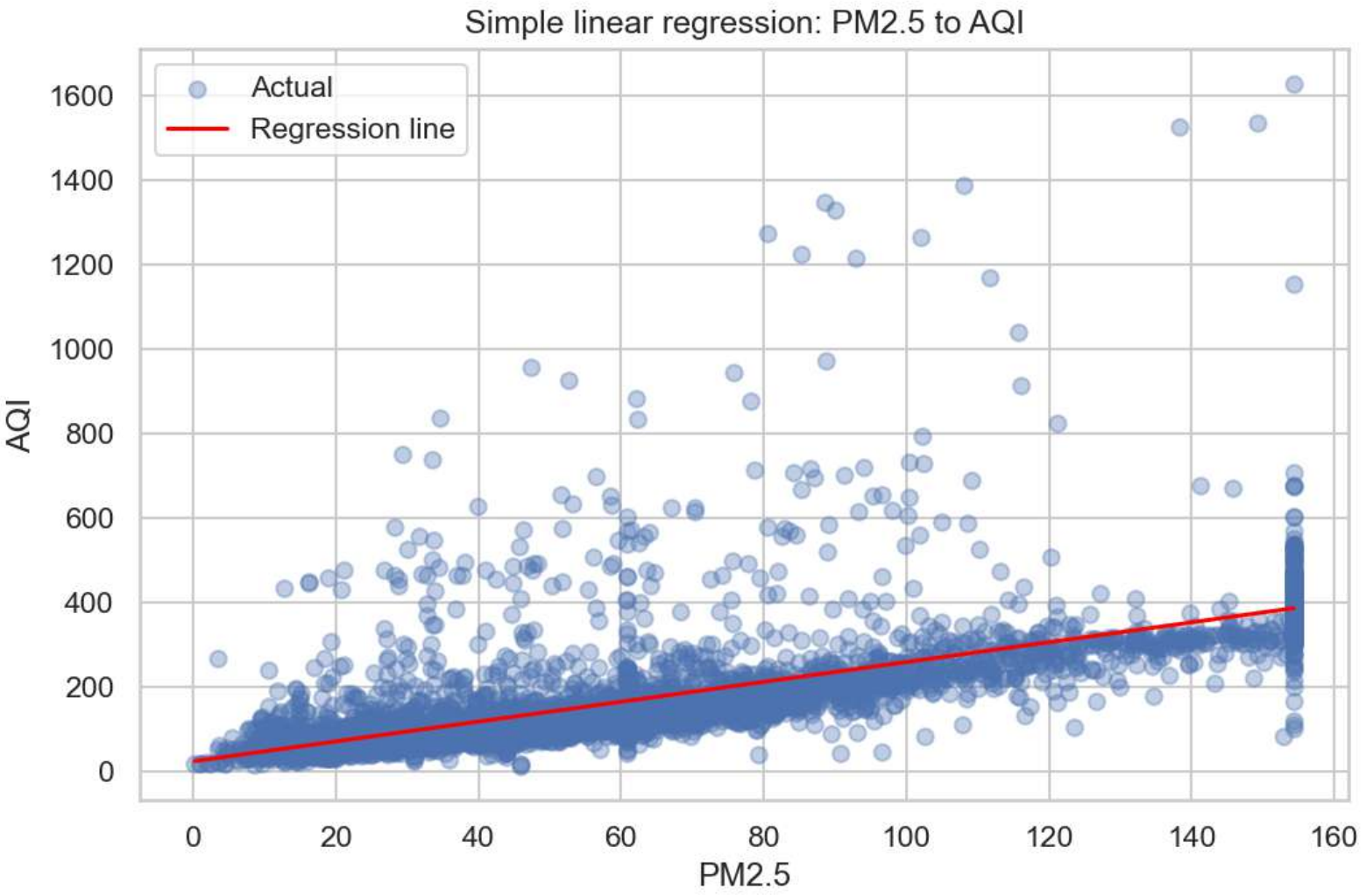

Pollutant-wise correlation with AQI
CO
PM2.5
NO2
PM10
SO2
NOx
NO
Toluene
O3
NH3
Benzene
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Correlation

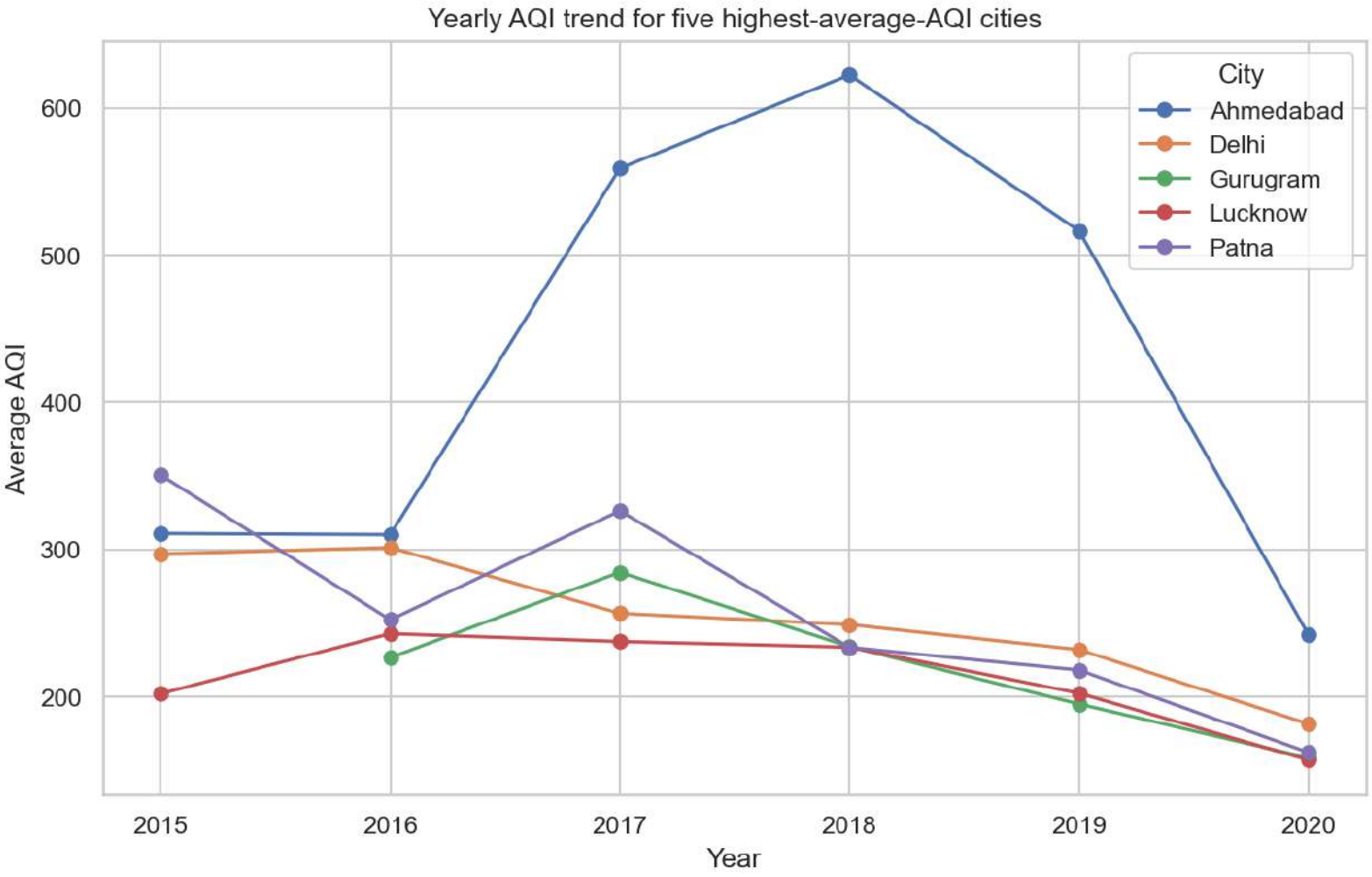


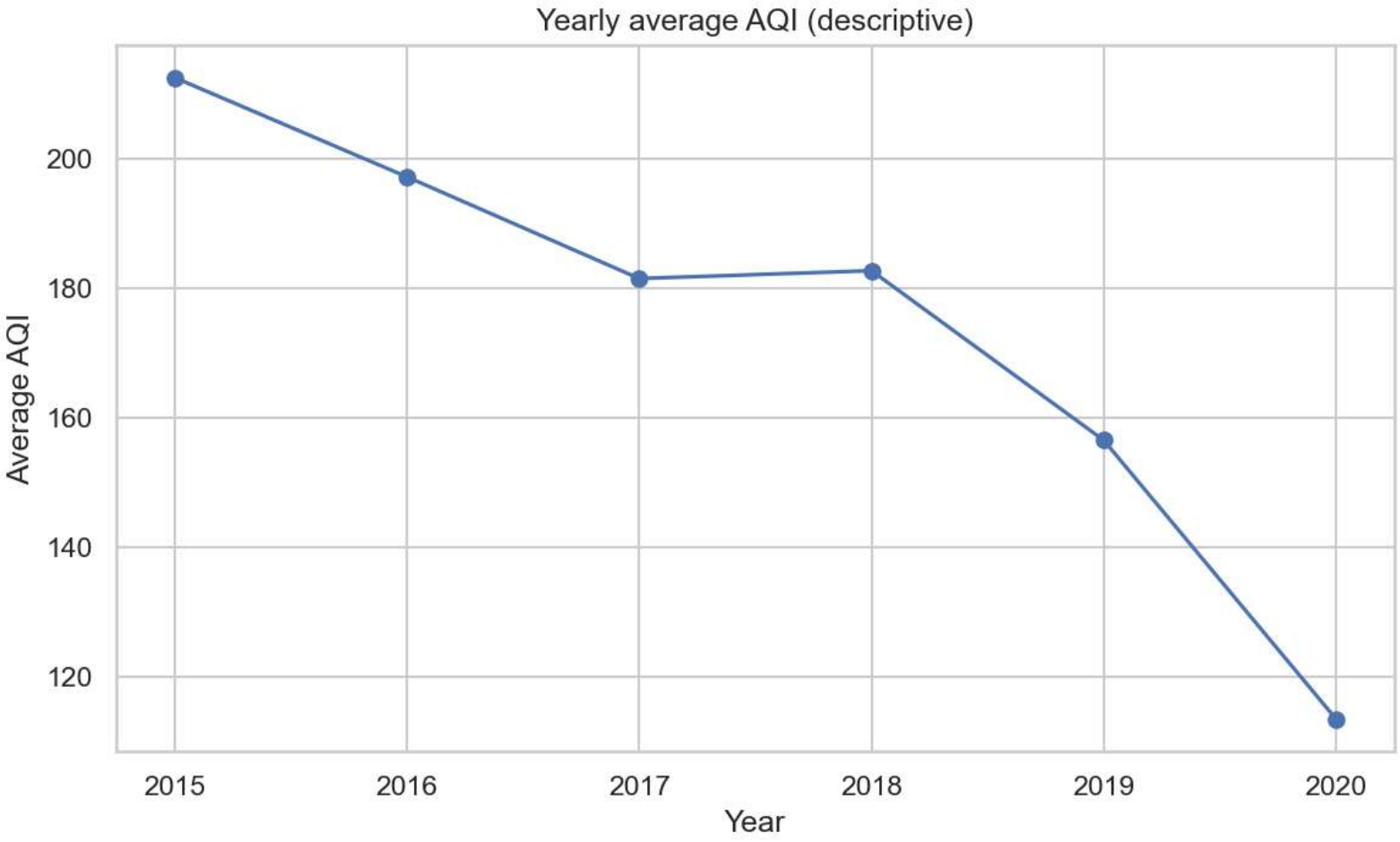
Model outlier limits are calculated from training data only.
The full cleaned dataset is not globally capped before splitting.


Multiple Regression: Actual vs Predicted AQI
Predicted AQI
Actual AQI
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600


Multiple regression residuals
Residual (actual - predicted)
1000
800
600
400
200
0
−200
0
100
200
300
400
500
600
Predicted AQI


Simple linear regression: PM2.5 to AQI
Actual
Regression line
AQI
PM2.5
0
200
400
600
800
1000
1200
1400
1600
0
20
40
60
80
100
120
140
160


Yearly AQI trend for five highest-average-AQI cities
City
Ahmedabad
Delhi
Gurugram
Lucknow
Patna
Average AQI
600
500
400
300
200
2015
2016
2017
2018
2019
2020
Year


Yearly average AQI (descriptive)
Average AQI
200
180
160
140
120
2015
2016
2017
2018
2019
2020
Year